

LECTURE 1

Spt 29th, 2025

DEF: (Norm) A function $\|\cdot\|: V \rightarrow \mathbb{R}$ s.t.

(i) $\|v\| \geq 0 \quad \forall v \in V$ and $\|v\| = 0 \Leftrightarrow v = 0$ definiteness

(ii) $\|av\| = |a| \|v\| \quad \forall v \in V, a \in \mathbb{R}$

(iii) $\|v+w\| \leq \|v\| + \|w\| \quad \forall w, v \in V$

CONVENTION: $x \in \mathbb{R}^n$ is treated as column vector

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

EXAMPLES (VECTOR NORMS): For $x \in \mathbb{R}^n$

• l^2 NORM: $\|x\|_2 := \sqrt{\sum_{i=1}^n |x_i|^2}$ (EUCLIDEAN NORM)

• l^p NORM: $\|x\|_p := \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \quad 1 \leq p < \infty$

• l^∞ NORM: $\|x\|_\infty := \max_{1 \leq i \leq n} |x_i| = \lim_{p \rightarrow \infty} \|x\|_p$

• For $w = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \in \mathbb{R}_{>0}$, define

$\|x\|_{p,w} := \left(\sum_{i=1}^n w_i |x_i|^p \right)^{1/p}$ (WEIGHTED p -NORM)

In particular, $p=2$

$$\|x\|_{2,w} = \sqrt{\sum_{i=1}^n w_i |x_i|^2} = \sqrt{x^T \underbrace{W}_{W = \text{diag}(w_1, \dots, w_n)} x}$$

• MAHALANOBIS NORM: for $A \in \mathbb{R}^{n \times n}$ positive definite, $\|x\|_A := \sqrt{x^T A x}$ is a norm.

$x^T = (x_1 \dots x_n)$ row vector

- MATRIX NORMS ← These are now fcts $\|\cdot\|: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$
 $\mathbb{R}^{m \times n} \ni X = \begin{bmatrix} x_{11} & \dots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \dots & x_{mn} \end{bmatrix}$

EXAMPLES (MATRIX NORMS) For $X \in \mathbb{R}^{m \times n}$

- $\|X\|_{H,p} := \left(\sum_{i=1}^m \sum_{j=1}^n |x_{ij}|^p \right)^{1/p}$ (HÖLDER p -NORM)

WARNING: Hölder ∞ -norm is not submultiplicative

$$\|X\|_{H,\infty} := \max_{1 \leq i \leq m} \max_{1 \leq j \leq n} |x_{ij}| \quad \text{But } X = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, Y = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$$

$$XY = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

- $\|X\|_F := \|X\|_{H,2} = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |x_{ij}|^2} = \sqrt{\text{tr}(X^T X)}$ (FROBENIUS NORM)

Frobenius norm is submultiplicative by Cauchy-Schwarz $\|XY\|_F \leq \|X\|_F \|Y\|_F$.
for Euclidean norm regarding the sum.

- Linear operator $X: (\mathbb{R}^n, \|\cdot\|_a) \rightarrow (\mathbb{R}^m, \|\cdot\|_b)$
 $v \mapsto Xv$

DEF: The operator norm (or induced norm) is defined as

$$\|X\|_{a,b} := \max_{v \neq 0} \frac{\|Xv\|_b}{\|v\|_a} = \max_{\|v\|_a=1} \|Xv\|_b = \max_{\|v\|_a \leq 1} \|Xv\|_b$$

(says how far v can get away from original position after X applied)

b/c $\|X\| = |\alpha| \|X\|$ and linearity of X

PROP: (Consistency Property) $\|Xv\|_b \leq \|X\|_{a,b} \|v\|_a \quad \forall v \in \mathbb{R}^n, X \in \mathbb{R}^{m \times n}$

• Caution: Not all norms are consistent

NOTE: Take $X \in \mathbb{R}^{m \times n}$ and $Y \in \mathbb{R}^{n \times p}$. Then $XY \in \mathbb{R}^{m \times p}$; i.e.,

$(\mathbb{R}^p, \|\cdot\|_c) \xrightarrow{Y} (\mathbb{R}^n, \|\cdot\|_b) \xrightarrow{X} (\mathbb{R}^m, \|\cdot\|_a)$. Then

$$\|XY\|_{a,c} \leq \|X\|_{a,b} \|Y\|_{b,c}.$$

DEF (MATRIX p,q -NORM) For $p, q \in [1, \infty]$ and $X \in \mathbb{R}^{m \times n}$, define

$$\|X\|_{p,q} := \max_{0 \neq v \in \mathbb{R}^n} \frac{\|Xv\|_q}{\|v\|_p}$$

Obs: $\|X\|_{1,\infty} = \|X\|_{\infty,1}$

• If $p=q$, we call this the MATRIX p -NORM.

EXAMPLES:

• MATRIX 1-NORM: $\|X\|_1 = \|X\|_{1,1} = \max_{1 \leq j \leq n} \sum_{i=1}^m |x_{ij}| = \text{max of } l^1 \text{ norms of columns of } X$

• MATRIX ∞ -NORM: $\|X\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |x_{ij}| = \text{max of } l^1 \text{ norms of rows of } X$

REMEMBER: " ∞ " looks like rows and
" 1 " looks like column

• MATRIX 2-NORM: $\|X\|_2 \stackrel{\text{def}}{=} \max_{0 \neq v \in \mathbb{R}^n} \frac{\|Xv\|_2}{\|v\|_2}$... no closed formula but we have facts about it

Can compute $p=1,2,\infty$
in polynomial time

= largest singular value of X

OBS: For $p \neq 1, 2, \infty$, NP-hard.

PF: (closed form formulas for $p=1, 2, \infty$ norms) **STUDY FOR TEST 2**

$$\bullet (\leq) \quad \|x\|_\infty \stackrel{\text{def}}{=} \max_{0 \neq y \in \mathbb{R}^n} \frac{\|xy\|_\infty}{\|y\|_\infty} = \max_{\|y\|_\infty=1} \|xy\|_\infty$$

$$= \max_{\|y\|_\infty=1} \left[\max_{1 \leq i \leq m} \left| \sum_{j=1}^n x_{ij} y_j \right| \right]$$

Triangle Inequality

$$\leq \max_{\|y\|_\infty=1} \left[\max_{1 \leq i \leq m} \sum_{j=1}^n |x_{ij}| |y_j| \right]$$

$$\max_{\|y\|_\infty=1} \Leftrightarrow y_1, \dots, y_n \leq 1 \rightarrow \leq \max_{1 \leq i \leq m} \sum_{j=1}^n |x_{ij}|$$

($\geq$) Need to find some $y \in \mathbb{R}^n$, $\|y\|_\infty=1$ such that this is satisfied

Take $y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n$ s.t. $y_j := \begin{cases} |x_{ij}|/x_{ij}, & \text{for } x_{ij} \neq 0 \\ 0, & \text{for } x_{ij} = 0 \end{cases} = \text{sgn}(x_{ij})$

Then $\|y\|_\infty = 1$ and $\|xy\|_\infty = \sum_{j=1}^n |x_{ij}|$

! LEARN THE OTHER ONES

LECTURE 2

Oct 4th, 2025

$$\lim_{n \rightarrow \infty} \|v_n\| = \left\| \lim_{n \rightarrow \infty} v_n \right\|$$

FACT: $\|\cdot\|: V \rightarrow \mathbb{R}$ is continuous. In fact, for $u, v \in V$,

$$\|u\| = \|u - v + v\| \leq \|u - v\| + \|v\|$$

$$\|v\| \leq \|u-v\| + \|u\|$$

$$\text{So, } \left| \|u\| - \|v\| \right| \leq \|u-v\| \Rightarrow \|\cdot\| \text{ is Lipschitz}$$

FACT: (EQUIVALENCE OF NORMS)

THM: For any finite dimensional space, all norms are equivalent.

DEF: $\|\cdot\|_a: V \rightarrow \mathbb{R}$ and $\|\cdot\|_b: V \rightarrow \mathbb{R}$ are equivalent

iff $\exists c_1, c_2$ s.t. $0 < c_1 < c_2$ and

$$c_1 \|v\|_a \leq \|v\|_b \leq c_2 \|v\|_a \quad \forall v \in V.$$

COR: If $v_k \rightarrow v$ in V (dim $V < \infty$) in $\|\cdot\|_a$, then $v_k \rightarrow v$ in $\|\cdot\|_b$.

NOTE: $v_k \rightarrow v$ in $\|\cdot\|_a \Leftrightarrow \|v_k - v\|_a \rightarrow 0$ by equivalence of norm and the squeeze then,

$$\begin{array}{ccc} c_1 \|v - v_k\|_a \leq \|v_k - v\|_b \leq c_2 \|v_k - v\|_a \\ \downarrow \qquad \qquad \qquad \downarrow \\ 0 \qquad \qquad \qquad \text{i.e., } \|v_k - v\|_b \rightarrow 0 \end{array}$$

EXAMPLE: Take $\mathbb{R}^n$ with $\|\cdot\|_2$ and $\|\cdot\|_\infty$. Then

$$\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty \quad (c_1 = 1, c_2 = \sqrt{n})$$

Pf: $\|x\|_2^2 = \sum_{i=1}^n x_i^2$, $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$. Say $\|x\|_\infty = |x_n|$ for some $n \in \{1, \dots, n\}$

So, since x_n is the largest entry,

$$x_n^2 \leq x_1^2 + \dots + x_n^2 \leq nx_n^2$$

$$\text{i.e., } \|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty.$$

NOTE: $\frac{1}{n} \|x\|_1 \leq \|x\|_\infty \leq \|x\|_1$ **PROVE THESE !**

EXAMPLE: For $\|\cdot\|_F$ and $\|\cdot\|_{H, \infty}$ in $\mathbb{R}^{m \times n}$, we have

$$\|X\|_{H, \infty} \leq \|X\|_F \leq \sqrt{mn} \|X\|_{H, \infty}$$

• For $\|X\|_\infty = \max \{l^1 \text{ norm of rows}\}$ and $\|X\|_2 = \max_{y \neq 0} \frac{\|Xy\|_2}{\|y\|_2}$.

$$\frac{1}{\sqrt{m}} \|X\|_2 \leq \|X\|_\infty \leq \sqrt{n} \|X\|_2$$

• Note

$$\|X\|_2 \leq \|X\|_F \leq \sqrt{n} \|X\|_2$$

INNER PRODUCTS

DEF: $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$ such that

(i) $\langle v, v \rangle \geq 0 \quad \forall v \in V$ (non-negativity)

(ii) $\langle v, w \rangle = 0 \quad \forall w \in V \iff v = 0$ (non-degeneracy)

(iii) $\langle \alpha v + \beta w, \gamma x + \delta y \rangle = \alpha \gamma \langle v, x \rangle + \alpha \delta \langle v, y \rangle + \beta \gamma \langle w, x \rangle + \beta \delta \langle w, y \rangle$
 $\forall v, w, x, y \in V$ and $\alpha, \beta, \gamma, \delta \in \mathbb{R}$. (bilinear)

(iv) $\langle v, w \rangle = \langle w, v \rangle \quad \forall v, w \in V$ (symmetry)

DEF: For an inner product $\langle \cdot, \cdot \rangle: V \times V \rightarrow \mathbb{R}$, define the induced norm by the inner product as

$$\|v\| := \sqrt{\langle v, v \rangle} \quad \forall v \in V.$$

THM: (CAUCHY-SCHWARZ) If $\|\cdot\|$ is induced by $\langle \cdot, \cdot \rangle$, then

$$|\langle v, w \rangle| \leq \|v\| \|w\| \quad \forall v, w \in V.$$

PROP: (PARALLELOGRAM IDENTITY) If for all $v, w \in (V, \|\cdot\|)$ we have

$$\|v+w\|^2 + \|v-w\|^2 = 2\|v\|^2 + 2\|w\|^2,$$

then $\|\cdot\|$ is induced by an inner product.

PROVE THIS

EXAMPLE: For $x, y \in \mathbb{R}^n$, the Euclidean inner product is $\langle x, y \rangle = x^T y$

• More generally, for $A \in \mathbb{R}^{n \times n}$ positive definite,

$$\langle x, y \rangle_A := \langle x, Ay \rangle = x^T A y$$

is an inner product.

NOTE: The Euclidean inner product induces the 2-norm

$$\sqrt{\langle x, x \rangle} = \sqrt{x^T x} = \|x\|_2$$

More generally, $\sqrt{\langle x, x \rangle_A} \stackrel{\text{def}}{=} \sqrt{\langle x, Ax \rangle} = \|x\|_A$ (Mahalanobis norm)

OBS: $\|\cdot\|_\infty$ is not induced by inner product.

EXAMPLE: (Matrix Inner Product) Take $\mathbb{R}^{m \times n}$. Then

$$\begin{aligned} \langle X, Y \rangle &:= \sum_{i=1}^m \sum_{j=1}^n x_{ij} y_{ij} \quad (\text{Standard inner product on } \mathbb{R}^{mn}) \\ &= \text{tr}(X^T Y) \end{aligned}$$

NOTE: $\sqrt{\langle X, X \rangle} = \sqrt{\text{tr}(X^T X)} = \sqrt{\sum_{i=1}^m \sum_{j=1}^n x_{ij}^2} \stackrel{\text{def}}{=} \|X\|_F$.

Cauchy - Schwarz: $|\text{tr}(X^T Y)| \leq \|X\|_F \|Y\|_F$.

————— " —————

MATRIX PRODUCTS

1) Inner Product: $x^T y \in \mathbb{R}$

2) Outer Product: $x y^T = \begin{pmatrix} x_1 y_1 & \dots & x_1 y_n \\ \vdots & \ddots & \vdots \\ x_m y_1 & \dots & x_m y_n \end{pmatrix}_{m \times n} \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$.
Always rank=1

3) Matrix - Matrix Product: $X \in \mathbb{R}^{m \times n}$, $Y \in \mathbb{R}^{n \times p}$

$$X = \begin{pmatrix} x_1^T \\ \vdots \\ x_m^T \end{pmatrix}, \quad Y = (y_1 \dots y_p) \rightsquigarrow XY = \begin{pmatrix} x_1^T y_1 & \dots & x_1^T y_p \\ \vdots & \ddots & \vdots \\ x_m^T y_1 & \dots & x_m^T y_p \end{pmatrix} \in \mathbb{R}^{m \times p}.$$

4) Take $X \in \mathbb{R}^{m \times n}$, $Y \in \mathbb{R}^{n \times p}$ and write

$$X = (x_1 \cdots x_n), \quad Y = \begin{pmatrix} y_1^T \\ \vdots \\ y_n^T \end{pmatrix} \rightsquigarrow XY = x_1 y_1^T + x_2 y_2^T + \cdots + x_n y_n^T \in \mathbb{R}^{m \times p}.$$

Matrix product is a sum of rank-1 matrices.

5) Composition of maps: $X: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $X \in \mathbb{R}^{m \times n}$ and $Y = (y_1 \cdots y_p) \in \mathbb{R}^{n \times p}$. Then $XY = (Xy_1 \cdots Xy_p)$.

6) Dual composition: $X = \begin{pmatrix} x_1^T \\ \vdots \\ x_n^T \end{pmatrix} \in \mathbb{R}^{m \times n}$, $XY = \begin{pmatrix} x_1^T Y \\ \vdots \\ x_n^T Y \end{pmatrix}$.

SPECIAL CASES:

(A) $X = (x_1 \cdots x_n) \in \mathbb{R}^{m \times n}$ and $Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n$. Then

$$XY = y_1 x_1 + \cdots + y_n x_n. \quad \leftarrow \text{This means } \text{im } X = \text{col } X.$$

(A') $x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$ and $Y = \begin{pmatrix} y_1^T \\ \vdots \\ y_n^T \end{pmatrix} \in \mathbb{R}^{m \times n}$. Then

$$x^T Y = x_1 y_1^T + \cdots + x_n y_n^T.$$

(B) $D = \text{diag}(d_1, \dots, d_n) \in \mathbb{R}^{n \times n}$, $X \in \mathbb{R}^{m \times n}$, $Y \in \mathbb{R}^{n \times m}$

$$DX = \begin{pmatrix} d_1 x_1^T \\ \vdots \\ d_n x_n^T \end{pmatrix} \quad \text{and} \quad YD = (d_1 y_1 \cdots d_n y_n).$$

↑
scales rows

↑
scales columns

REMARK: $\|X\|_1 = \max\{\ell^1 \text{ norm of cols. of } X\}$

$$\begin{aligned} X &= (x_1 \dots x_n) \quad \|X\|_1 = \max_{\|y\|_1=1} \|Xy\|_1 = \max_{\|y\|_1=1} \|y_1 \vec{x}_1 + \dots + y_n \vec{x}_n\|_1 \\ &\leq \max_{\|y\|_1=1} (|y_1| \|\vec{x}_1\|_1 + \dots + |y_n| \|\vec{x}_n\|_1) \\ &= \left[\max_{\|y\|_1=1} (|y_1| + \dots + |y_n|) \right] \left[\max(\|x_1\|_1, \dots, \|x_n\|_1) \right] \end{aligned}$$

LECTURE 3

EIGENVALUES & EIGENVECTORS

Oct 6th, 2025

EIGENPROBLEM: $Ax = \lambda x$, $A \in \mathbb{R}^{n \times n}$, $x \in \mathbb{C}^n \setminus \{0\}$, $\lambda \in \mathbb{C}$
"right eigenvector"

• CONVENTION: Normalize eigenvectors to have $\|x\| = 1$.

DEF: (Eigenspace) $E_\lambda := \{x \in \mathbb{C}^n : Ax = \lambda x\}$ $\leftarrow$ set of all eigenvectors together with zero vector.

(Spectrum) $\lambda(A) := \{\lambda \in \mathbb{C} : Ax = \lambda x \text{ for some } x \in \mathbb{C}^n \setminus \{0\}\}$
 $= \{\lambda_1, \dots, \lambda_n\}$ (SPECTRUM)
not necessarily distinct

CONVENTION: order of eigenvalues $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$.

$\lambda_1 \equiv \lambda_{\max} \equiv$ principal eigenvalue; $|\lambda_1| =: \rho(A)$ SPECTRAL RADIUS

Obs: Any $A \in \mathbb{C}^{n \times n}$ has n eigenvalues counted with multiplicity

|||
ALGEBRAIC
MULTIPLICITY

LEFT EIGENVECTORS: $y^T A = \lambda y^T$, $y \in \mathbb{C}^n \setminus \{0\}$

Transpose ↙

$$A^T y = \lambda y$$

i.e., left-eigenvectors of A are right-eigenvectors of A^T .

NOTE: Left and right eigenvectors are usually distinct, but right and left eigenvalues are the same. (EW's of A are roots of $0 = \det(A - \lambda I_n) = \det(A^T - \lambda I_n)$
 $\Rightarrow \sigma(A) = \sigma(A^T)$ since $\det(X) = \det(X^T)$.)

COR: $\lambda(A) = \lambda(A^T)$.

DIAGONALIZATION $A \in \mathbb{C}^{n \times n}$ with $\bigvee$ (lin. indep.) eigenvectors $x_1, \dots, x_n \in \mathbb{C}^n$ and eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{C}$. Take

$$X = \begin{pmatrix} | & & | \\ x_1 & \dots & x_n \\ | & & | \end{pmatrix} \in \mathbb{C}^{n \times n}$$

Then

$$AX = \begin{pmatrix} | & & | \\ \lambda_1 x_1 & \dots & \lambda_n x_n \\ | & & | \end{pmatrix} = X \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} =: X\Lambda$$

i.e., since the cols. of X are lin. indep., it's invertible:

$$\boxed{A = X \Lambda X^{-1}}$$

$\iff$

$$\boxed{X^{-1} A X = \Lambda}$$

(EVD
Eigenvalue
Decomposition)

NOTE: We say A IS DIAGONALIZABLE.

DEF: # of lin. indep. eigenvectors $\equiv$ GEOMETRIC MULTIPLICITY.

SPECTRAL THEOREMS

DEF: (Normal Matrix) $A \in \mathbb{C}^{n \times n}$ is normal iff $A^*A = AA^*$.

A^* = conjugate transpose = $[\overline{a_{ji}}]_{i,j=1,\dots,n}$.

THM: (Complex Spectral Theorem) For $A \in \mathbb{C}^{n \times n}$, TFAE

(i) A is normal; i.e., $A^*A = AA^*$

(ii) $\mathbb{C}^n$ has an orthonormal basis consisting of eigenvectors of A

(iii) There exists $Q \in \mathbb{C}^{n \times n}$, $Q^*Q = I$, such that ($Q \in U(n)$)

$$A = Q\Lambda Q^* \Leftrightarrow \Lambda = Q^*AQ$$

and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_j \in \mathbb{C}$.

COR: (Spectral Theorem for Hermitian matrices) If $A \in \mathbb{C}^{n \times n}$ is Hermitian (i.e., self-adjoint $\Leftrightarrow A^* = A$), then there exists a unitary $Q \in \mathbb{C}^{n \times n}$, $Q^*Q = I$, and

$$A = Q\Lambda Q^* \Leftrightarrow \Lambda = Q^*AQ = \text{diag}(\underbrace{\lambda_1, \dots, \lambda_n}_{\substack{\uparrow \\ \mathbb{R}}})$$

REMARK: Self-adjoint (i.e., Hermitian) operators have real eigenvalues.

COR: (Spectral Theorem for Symmetric Matrices) If $A \in \mathbb{R}^{n \times n}$ is symmetric (i.e., $A^T = A$), then $\exists Q \in \mathbb{R}^{n \times n}$ orthogonal (i.e., $Q^T Q = Q Q^T = \mathbb{1}$) such that

$$A = Q \Lambda Q^T \iff \Lambda = Q^T A Q.$$

DEF: $U(n) := \{Q \in \mathbb{C}^{n \times n} : Q^* Q = Q Q^* = \mathbb{1}\}$ UNITARY

$O(n) := \{Q \in \mathbb{R}^{n \times n} : Q^T Q = Q Q^T = \mathbb{1}\}$ ORTHOGONAL

$SU(n) := \{Q \in U(n) : \det Q = 1\}$ SPECIAL UNITARY

$SO(n) := \{Q \in O(n) : \det Q = 1\}$ SPECIAL ORTHOGONAL

$GL(n) = \{A \in \mathbb{C}^{n \times n} : A \text{ invertible}\}$ GENERAL LINEAR

JORDAN CANONICAL FORM

$$A \in \mathbb{C}^{n \times n}$$

$$m_j \in \{0, 1\}$$

$$A = X \begin{bmatrix} \lambda_1 & 1 & & 0 \\ & \lambda_1 & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda_n \end{bmatrix} X^{-1}$$

Useless because ↴

• Golub-Wilkinson Thm:
JCF cannot be computed in finite precision.

Can prove stuff with it though...

Obs: $A^k = (X J X^{-1})^k = X J^k X^{-1}$ which is nice.

SPECTRAL RADII

DEF: The spectral radius of $A \in \mathbb{C}^{n \times n}$ is

$$\rho(A) := \max \{ |\lambda| : Ax = \lambda x \text{ for some } x \in \mathbb{C}^n \setminus \{0\} \}$$
$$= |\lambda_{\max}| = |\lambda_1|.$$

Gelfand's Formula:

$$\rho(A) = \lim_{k \rightarrow \infty} \|A^k\|^{1/k}$$

LEMMA: If $\|\cdot\|: \mathbb{C}^{n \times n} \rightarrow \mathbb{R}$ is consistent (i.e., $\|Ax\| \leq \|A\| \|x\|$) then $\rho(A) \leq \|A\|$ (true for any consistent norm)

Pf: Let $x \in \mathbb{C}^n$ be eigenvect. associated w/ $\lambda_{\max}$, then

$$\|A\| = \|A\| \|x\| \geq \|Ax\| = \|\lambda_{\max} x\| = |\lambda_{\max}| \|x\|$$

$\uparrow$
consistency

$$= \rho(A) \|x\|$$

THM: For any $\varepsilon > 0$, there exists some operator norm

$$\|A\|_\alpha = \max_{x \neq 0} \frac{\|Ax\|_\alpha}{\|x\|_\alpha}, \quad \|\cdot\|_\alpha: \mathbb{C}^n \rightarrow \mathbb{R},$$

such that

lemma above b/c all operator norms are consistent

$$\rho(A) \leq \|A\|_\alpha \leq \rho(A) + \varepsilon.$$

$\uparrow$
JCF

LEMMA: For $A \in \mathbb{C}^{n \times n}$, $A^k \xrightarrow{k \rightarrow \infty} 0$ iff $\rho(A) < 1$

PROVE THIS

LECTURE 4

Oct 8th, 2025

Find estimates as to where the eigenvalues of a matrix are

GERSHGORIN THEOREM

THM: (Gershgorin) Let $A = (a_{ij})_{i,j=1}^n \in \mathbb{C}^{n \times n}$ and define the Gershgorin disks as

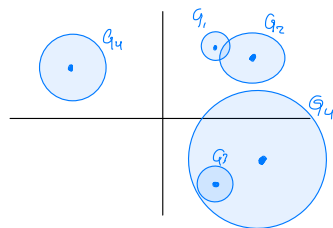
$$G_i = B_{r_i}(a_{ii}) \quad i = 1, \dots, n,$$

ball of radius r_i centered at a_{ii}

where $r_i := \sum_{j \neq i} |a_{ij}|$.

Then,

$$(1) \quad \text{Spec}(A) \subset \bigcup_{i=1}^n G_i.$$



(2) The # of eigenvalues of A contained in any (connected) component of $\bigcup_{i=1}^n G_i$ is the # of G_i 's in that component.

DEF: A matrix $A \in \mathbb{C}^{n \times n}$ is strictly diagonally dominant iff

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}| \quad \forall i = 1, \dots, n.$$

LEMMA: Every strictly diagonally dominant matrix is invertible.

PF: Say $A \in \mathbb{C}^{n \times n}$ is SSD and, by contradiction, $A \notin GL(n, \mathbb{C})$.

Then $Ax = 0$ for some $x \neq 0$. Denote $|x_k| := \|x\|_\infty > 0$.
Look at k th row of Ax :

$$0 = \sum_{j=1}^n a_{kj} x_j \quad \text{i.e., } a_{kk} x_k = - \sum_{j \neq k} a_{kj} x_j$$

So,

$$|a_{kk}| |x_k| = |a_{kk} x_k| \leq \sum_{j \neq k} |a_{kj} x_j| = \sum_{j \neq k} |a_{kj}| |x_j|$$

$$\|x\|_\infty = |x_k| \leq |x_k| \sum_{j \neq k} |a_{kj}|$$

$$\text{i.e., } |a_{kk}| \leq \sum_{j \neq k} |a_{kj}|$$

↔ // ↔
strictly diagonally
dominant

Thus, A is invertible.

STUDY THIS PROOF !
FOR MIDTERM !

PF: (Gershgorin Thm) Let $z \notin \bigcup_{i=1}^n G_i$. We want to show that this implies $z \notin \text{Spec}(A)$. But note that $(A - zI)$ is invertible by the definition of Gershgorin disks & lemma above.

MIDTERM: PROVE GESZTORIN THM $\Rightarrow$ LEMMA

REMARK: $p(z) = x^d + c_{d-1}x^{d-1} + \dots + c_0 = \det(x\mathbb{1} - C)$, where

$$C = \begin{bmatrix} 0 & & & -c_0 \\ 1 & 0 & & -c_1 \\ & 1 & \ddots & \vdots \\ 0 & & \ddots & 0 \\ & & & 1 & -c_{d-1} \end{bmatrix}$$

QR-DECOMPOSITION (i.e., Gram-Schmidt)

For $A \in \mathbb{C}^{n \times n}$, there exists $Q \in U(n)$, $R = \begin{bmatrix} r_{11} & & * \\ & \ddots & \\ & & r_{nn} \end{bmatrix} \in \mathbb{C}^{n \times n}$
upper triangular s.t. $A = QR$

• FRANCIS QR-ALGORITHM: Use QR-decomposition to find the eigenvalues of a matrix

Set: $A_0 = A$

Step 1: $A_0 = Q_0 R_0$

Step 2: $R_0 Q_0 = A_1$

Step 3: $A_1 = Q_1 R_1$

Step 4: $R_1 Q_1 = A_2$

$\vdots$

NOTE: $A_1 = R_0 Q_0 = Q_0^* A_0 Q_0$ and conjugation preserves the spectrum b/c it's a similarity transformation.

CLAIM: $Q_k \xrightarrow{k \rightarrow \infty} Q_\star$ } $U(n)$ is compact hence converges
 $R_k \xrightarrow{k \rightarrow \infty} R_\star$ } upper triangular

THEN: $A := Q_{\star} R_{\star} Q_{\star}^* = Q_{\star} \begin{bmatrix} \tilde{r}_{11} & & & \\ & \tilde{r}_{22} & & \\ & & \ddots & \\ & & & \tilde{r}_{nn} \end{bmatrix} Q_{\star}^*$
 (SCHUR DECOMPOSITION)

NOTE: $\text{Spec}(A) = \text{Spec}(R_{\star}) = \{\tilde{r}_{11}, \dots, \tilde{r}_{nn}\}$.

SINGULAR VALUE DECOMPOSITION (SVD) ← ALWAYS POSSIBLE !

For a matrix $A \in \mathbb{R}^{n \times n}$, the SVD is

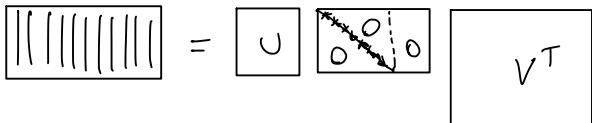
$$A = U \Sigma V^T,$$

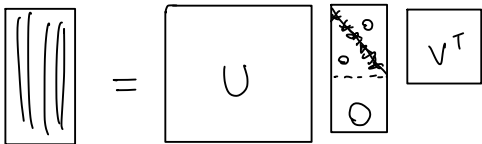
← if $A \in \mathbb{C}^{n \times n}$, switch "T" → "H"

$U \in \mathbb{R}^{n \times n}$ unitary, $V \in \mathbb{R}^{n \times n}$ unitary, $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$,

$r := \text{rank } A$.

For non-square matrix $A \in \mathbb{R}^{m \times n}$,

($m < n$) short and fat: 

($n < m$) Tall and skinny: 

DEF: columns of U → left singular vectors
 columns of V → right singular vectors
 diagonal entries of Σ → singular values

Singular Values = Eigenvalues of $\sqrt{A^* A}$, i.e., $\sqrt{\text{eigenvals of } A^* A}$
 $\Updownarrow$
 $\sqrt{A A^*}$

THM: (SVD) Given any $A \in \mathbb{R}^{m \times n}$, there exists $U \in O(m)$, and $V \in O(n)$, and

$$\Sigma := \left[\begin{array}{c|c} \sigma_1 & 0 \\ \vdots & \\ \hline 0 & 0 \end{array} \right] \in \mathbb{R}^{m \times n},$$

such that

$$A = U \Sigma V^T.$$

Moreover, $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$, $r := \text{rank } A$.

REMARK: We can write.

$$\begin{aligned} A &= U \Sigma V^T = \sigma_1 u_1 v_1^T + \dots + \sigma_r u_r v_r^T + 0 + 0 + \dots = \sum_{j=1}^r \sigma_j u_j v_j^T. \\ &= \begin{bmatrix} | & & | \\ u_1 & \dots & u_r \\ | & & | \end{bmatrix}_{m \times r} \begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r \end{bmatrix}_{r \times r} \begin{bmatrix} | & & | \\ v_1 & \dots & v_r \\ | & & | \end{bmatrix}_{r \times n} \\ &=: U_r \Sigma_r V_r^T \quad (\text{CONDENSED SVD} \leftarrow \text{does same info.}) \end{aligned}$$

PF: (Condensed SVD) $A \in \mathbb{R}^{m \times n}$.

$$W = \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \in \mathbb{R}^{(m+n) \times (m+n)}$$

$\iff W^T = W$ thus it is diagonalizable by Spectral Theorem.

$$W = Z \Lambda Z^{-1}$$

Say $Wz = \sigma z$, $z = \begin{bmatrix} x \\ y \end{bmatrix}$, $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$.

$\uparrow$
Moreover, eigenvalues of W are real.

Then

$$\begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \sigma \begin{bmatrix} x \\ y \end{bmatrix} \rightsquigarrow \begin{cases} Ay = \sigma x \\ A^T x = \sigma y \end{cases}.$$

$$\begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} x \\ -y \end{bmatrix} = \begin{bmatrix} -Ay \\ A^T x \end{bmatrix} = \begin{bmatrix} -\sigma x \\ \sigma y \end{bmatrix} = -\sigma \begin{bmatrix} x \\ -y \end{bmatrix}.$$

So,

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 = 0 \dots 0 > -\sigma_r \geq -\sigma_{r-1} \geq \dots \geq -\sigma_1$$

$$\text{WLOG: } \|z\|^2 = 2 \iff \|z\| = \sqrt{2}$$

$$\begin{aligned} & \parallel \\ z^T z &= \begin{bmatrix} x^T & y^T \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x^T x + y^T y. \end{aligned}$$

Thus, $\begin{bmatrix} x \\ y \end{bmatrix}^T \begin{bmatrix} x \\ -y \end{bmatrix} = 0$ b/c they are eigenvectors corresponding to diff. eigenvalues.

$$\parallel \\ x^T x - y^T y$$

$$\text{So: } \begin{cases} x^T x + y^T y = 2 \\ x^T x - y^T y = 0 \end{cases} \rightsquigarrow x^T x = 1 = y^T y \text{ (only solution)}$$

Let $X := [x_1, \dots, x_r]$, $Y = [y_1, \dots, y_r]$. Then

$$Z = \begin{bmatrix} X & X \\ Y & -Y \end{bmatrix}.$$

LECTURE 5

Oct 13th, 2025

SINGULAR VALUE DECOMPOSITION

$A \in \mathbb{R}^{m \times n}$ can be decomposed into

$$A = U \Sigma V^T = \begin{bmatrix} | & & | \\ u_1 & \cdots & u_m \\ | & & | \end{bmatrix} \begin{bmatrix} \sigma_1 & \cdots & \sigma_r & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 0 & 0 \end{bmatrix} \begin{bmatrix} - & v_1^T & - \\ \vdots & \vdots & \vdots \\ - & v_n^T & - \end{bmatrix}$$

$r = \text{rank}(A)$

$$=: \begin{bmatrix} U_r & | & U_{m-r} \end{bmatrix} \begin{bmatrix} \Sigma_r & | & 0 \\ \hline 0 & | & 0 \end{bmatrix} \begin{bmatrix} - & V_r^T & - \\ \vdots & \vdots & \vdots \\ - & V_{n-r}^T & - \end{bmatrix}$$

- Condensed SVD: $A = U_r \Sigma_r V_r^T$.

PROPERTIES OF SVD:

(1) $\dim \text{im } A = \text{rank}(A) = r = \# \text{ nonzero singular values}$

$$\dim \ker A = n - r = \text{nullity}(A)$$

(2) columns of U_r = orthonormal basis for $\text{im}(A)$

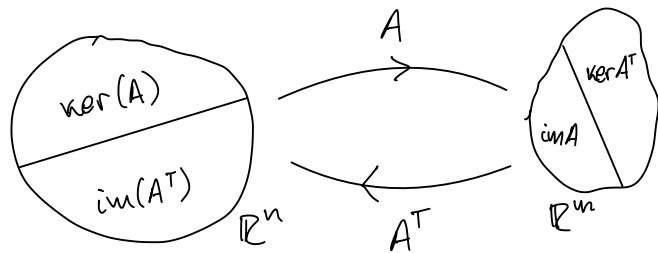
columns of U_{n-r} = orthonormal basis for $\ker(A)$

columns of V_r = orthonormal basis for $\text{im}(A^T)$

columns of V_{n-r} = orthonormal basis for $\ker(A^T)$

(3) FREDHOLM'S ALTERNATIVE

$$\begin{cases} \mathbb{R}^m = \text{im } A \oplus \ker A^T \\ \mathbb{R}^n = \text{im } A^T \oplus \ker A \end{cases}$$



Projection Operators: Take $W \subset \mathbb{R}^n$, then any $P^2 = \mathbb{1}$ is a projection. If, in addition, $P^T = P$, then P is an orthogonal projection.

NOTE: $\text{proj}_{\text{im}(A)} = U_r U_r^T$; $\text{proj}_{\ker(A)} = U_{n-r} U_{n-r}^T$; $\text{proj}_{\text{im}(A^T)} = V_r V_r^T$;
 $\text{proj}_{\ker(A^T)} = U_{m-r} U_{m-r}^T$

SVD AND NORMS

- FROBENIUS NORM** $\|A\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_r^2} = \|(\sigma_1, \dots, \sigma_r)\|_2$
- MATRIX 2-NORM** $\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sigma_1 \leftarrow \text{LARGEST SINGULAR VALUE}$
 (AKA SPECTRAL NORM) $= \|(\sigma_1, \dots, \sigma_r)\|_\infty$
- NUCLEAR NORM** $\|A\|_* := \|(\sigma_1, \dots, \sigma_r)\|_1 = \sigma_1 + \dots + \sigma_r$
- SCHATTEN p-NORM** $\|A\|_{S_p} := \|(\sigma_1, \dots, \sigma_r)\|_p = \left(\sum_{j=1}^r \sigma_j^p \right)^{1/p}$
- KY FAN (p, κ)-NORM** $\|A\|_{S_{p,\kappa}} := \left(\sigma_1^p + \dots + \sigma_\kappa^p \right)^{1/p}, 1 \leq \kappa \leq r$

- If $m=n$, $|\det(A)| = \sigma_1 \cdots \sigma_n$
- All of these are unitarily invariant:
 $\|Q_1 A Q_2\| = \|A\|$ for any $Q_1 \in O(m)$, $Q_2 \in O(n)$.

———— // ————

SOLVING LINEAR SYSTEMS

GOAL: Given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, want to find $x \in \mathbb{R}^n$ s.t.

$$Ax = b.$$

Problems: existence $\leftrightarrow b \notin \text{im } A$
 uniqueness $\leftrightarrow \ker A \neq \{0\}$

CAVEAT: from finite precision "errors" or measurement errors, we have error corruption

$$\begin{array}{c} (A+E)x = b+r \\ \uparrow \qquad \qquad \uparrow \\ \text{ERROR} \qquad \text{ERROR} \end{array}$$

- $E \neq 0 \rightsquigarrow$ Total Least Squares
- $E = 0 \rightsquigarrow$ Ordinary Least Squares

ORDINARY LEAST SQUARES (OLS)

$$Ax \approx b \iff \min_{x \in \mathbb{R}^n} \|Ax - b\|_2^2 \quad \leftarrow \text{Use 2-norm here b/c it's differentiable}$$

NOTE: $\|\cdot\|_2$ is convex; thus there is always a solution

Using SVD:

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|_2^2 = \min_{x \in \mathbb{R}^n} \|U \Sigma V^T x - b\|_2^2$$

$$= \min_{x \in \mathbb{R}^n} \|U(\Sigma V^T x - U^T b)\|_2^2$$

$\|\cdot\|_2^2$ is unitarily invariant $\rightarrow$

$$= \min_{x \in \mathbb{R}^n} \|\Sigma V^T x - U^T b\|_2^2$$

set $y := V^T x$

$c := U^T b$

$$= \min_{y \in \mathbb{R}^n} \|\Sigma y - c\|_2^2$$

same as min because V^T is invertible in particular

$$= \min_{y \in \mathbb{R}^n} \left\| \begin{pmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r & 0 \\ & & & \ddots & \\ 0 & & 0 & & 0 \end{pmatrix} \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} - \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} \right\|_2^2$$

$$= \min_{y \in \mathbb{R}^n} \left\| \begin{pmatrix} \sigma_1 y_1 - c_1 \\ \vdots \\ \sigma_r y_r - c_r \\ -c_{r+1} \\ \vdots \\ -c_n \end{pmatrix} \right\|_2^2$$

$$= (\underbrace{\sigma_1 y_1 - c_1}_{\stackrel{!}{=} 0})^2 + \dots + (\underbrace{\sigma_r y_r - c_r}_{\stackrel{!}{=} 0})^2 + c_{r+1}^2 + \dots + c_n^2$$

i.e.,

$$y_j \stackrel{!}{=} \frac{c_j}{\sigma_j} \quad \forall j = 1, \dots, r$$

Upshot: $x \stackrel{\text{def}}{=} V y = V \begin{pmatrix} c_1/\sigma_1 \\ \vdots \\ c_r/\sigma_r \\ y_{r+1} \\ \vdots \\ y_n \end{pmatrix}, \quad y_{r+1}, \dots, y_n \in \mathbb{R}$

ALL SOLUTIONS!

Maybe not unique

MINIMUM NORM LEAST SQUARES $\leftarrow$ solves the question of uniqueness
(MNOLS)

$$\min \left\{ \|x\|_2 : x \in \arg \min_{x \in \mathbb{R}^n} \|Ax - b\|_2 \right\}$$

$\|$ $\leftarrow$ from previous page

$$\min \left\{ \|x\|_2 : x = V \begin{pmatrix} c_1/\sigma_1 \\ \vdots \\ c_r/\sigma_r \\ y_{r+1} \\ \vdots \\ y_n \end{pmatrix}, y_{r+1}, \dots, y_n \in \mathbb{R}^n \right\}$$

NOTE: $\|x\|_2^2 = \|Vy\|_2^2 = \|y\|_2^2$

$\|\cdot\|_2$ is unitarily invariant $= \left(\frac{c_1}{\sigma_1}\right)^2 + \dots + \left(\frac{c_r}{\sigma_r}\right)^2 + y_{r+1}^2 + \dots + y_n^2$

SOLUTION: Set $y_{r+1} = \dots = y_n \stackrel{!}{=} 0$. Then, the general solution to MNOLS is

$$x^+ := V \begin{pmatrix} c_1/\sigma_1 \\ \vdots \\ c_r/\sigma_r \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Recall

$$\begin{cases} c := U^T b \\ x := Vy \end{cases}$$

$$\text{and } y = \begin{pmatrix} c_1/\sigma_1 \\ \vdots \\ c_r/\sigma_r \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 1/\sigma_1 & & 0 \\ & \ddots & \\ 0 & & 1/\sigma_r & 0 & \dots & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_m \end{pmatrix}$$

$$\Sigma^+ := \text{diag}\left(\frac{1}{\sigma_1}, \dots, \frac{1}{\sigma_r}, 0, \dots, 0\right) \in \mathbb{R}^{n \times m}$$

Thus,

$$x^+ = V \left[\begin{array}{ccc|cc} 1/\sigma_1 & & & & \\ & \ddots & & & \\ & & 1/\sigma_r & & \\ \hline & & & 0 & \\ & 0 & & & 0 \end{array} \right] U^T b \quad =: \Sigma^+$$

$$(A^+ := V \Sigma^+ U^T)$$

!!
 A^+ (MOORE-PENROSE
 PSEUDO-INVERSE)



Upshot: Solution to OLS is $x^+ = A^+ b \stackrel{\text{def}}{=} V \Sigma^+ U^T b$

REMARK: When $A \in GL(n, \mathbb{R})$, then $A^+ = A^{-1}$
 ($m=n=r$)

→ Math prof (topology) at UChicago

THM: (Moore-Penrose) For any $A \in \mathbb{R}^{m \times n}$, there exists a unique matrix $X \in \mathbb{R}^{n \times m}$ such that

$$(1) \quad AXA = A$$

$$(2) \quad XAX = X$$

$$(3) \quad (AX)^T = AX$$

$$(4) \quad (XA)^T = XA$$

(This unique X here
 is given by
 $X = V \Sigma^+ U^T = A^+$)

↑
 PROVE THIS

LECTURE 6

Oct 15th, 2025

LEAST SQUARES: Given $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $\alpha > 0$, want to solve

!

$$\min_{\|x\| \leq \alpha} \|Ax - b\|.$$

REMARK: if $\alpha < \|A^T b\|$, this is CONSTRAINED MINIMIZATION and, in this case,

!

$$\min_{\|x\| \leq \alpha} \|Ax - b\| = \min_{\|x\| = \alpha} \|Ax - b\|$$

← Equality constraint minimization
↓
Lagrange Multiplier

!

LAGRANGIAN: $\mathcal{L}(x, \mu) = \|Ax - b\|^2 - \mu(\|x\|^2 - \alpha^2)$

!

$$0 \stackrel{!}{=} \nabla_x \mathcal{L}(x, \mu) = 2A^T(Ax - b) + 2\mu x$$

↕

$$(A^T A + \mu \mathbb{1})x = A^T b$$

$$\Rightarrow x = (A^T A + \mu \mathbb{1})^{-1} A^T b$$

Take SVD
 $A = U \Sigma V^T$

$$\alpha^2 = \|x\|^2 = \|(A^T A + \mu \mathbb{1})^{-1} A^T b\|^2$$

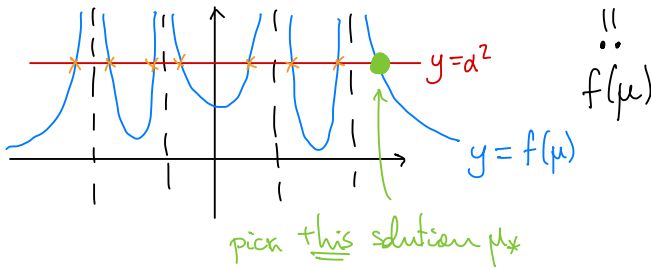
↪

$$c \stackrel{!}{=} U^T b \Rightarrow c^T \Sigma (\Sigma^T \Sigma + \mu \mathbb{1})^{-2} \Sigma^T c$$

STUDY LAGRANGE MULTIPLIERS

SEE PG 55

$$\Rightarrow \alpha^2 = \|x\|^2 = \sum_{i=1}^r \frac{c_i^2 \sigma_i^2}{(\sigma_i^2 + \mu)^2} \quad \left. \vphantom{\sum_{i=1}^r} \right\} \begin{array}{l} \text{SECULAR} \\ \text{EQUATION} \end{array}$$



- NEWTON - RAPHSON is method to find solutions to $f(\mu) = \alpha^2$.

————— //

MATRIX APPROXIMATION PROBLEM

Given $A \in \mathbb{R}^{n \times n}$, what, for example symmetric, matrix X is the closest to A in $\|\cdot\|_F$?

$$\Leftrightarrow \min_{X \in \text{Sym}(n, \mathbb{R})} \|A - X\|_F \quad \xrightarrow{?} \quad X = \frac{A^T + A}{2} \quad (\text{no SVD required})$$

$$\begin{aligned} \min_{X^T X = I} \|A - X\|_F & \quad \leftarrow \quad \|A - X\|_F^2 = \text{tr}[(A - X)^T (A - X)] \\ & = \text{tr}(AA^T - X^T A - A^T X + X^T X) \\ & = \cancel{\|A\|_F^2} - 2 \text{tr}(A^T X) + \cancel{\|X\|_F^2} \end{aligned}$$

$$\text{So } \min_{X^T X = I} \|A - X\|_F \Leftrightarrow -2 \max_{X^T X = I} \text{tr}(A^T X)$$

Now, $\max_{X^T X = I} \text{tr}(A^T X) \stackrel{A = U \Sigma V^T (\text{SVD})}{=} \max_{X^T X = I} \text{tr}(V \Sigma^T U^T X)$

$$= \max_{X^T X = I} \text{tr}(\Sigma^T \underbrace{U^T X V}_{\substack{=: Z \in O(n) \Rightarrow |z_{ij}| \leq 1 \\ \text{!}}})$$

$$\text{tr}(\Sigma^T Z) \leq \sigma_1 |z_{11}| + \dots + \sigma_n |z_{nn}|$$

$$\leq \sigma_1 + \dots + \sigma_n. \quad \max \Leftrightarrow Z = I \quad \Rightarrow X = UV^T$$

" $U^T X V$

$(Z \in O(n) \Rightarrow Z^T Z = I, \text{ hence } (Z^T Z)_{jj} = 1 \Rightarrow \sum_i z_{ij}^2 = 1 \Rightarrow |z_{ij}| \leq 1)$
i.e., $I = \sum_j (Z^T)_j (Z)_j = \sum_j z_{1j} z_{1j} = \sum_j z_{1j}^2 \geq \sum_j z_{1j}^2$

Upshot: $\min_{X^T X = I} \|A - X\|_F = UV^T, \text{ SVD of } A \text{ is } A = U \Sigma V^T$

Closest rank- r matrix: $\min_{\text{rank } X \leq r} \|A - X\|_F$

$\uparrow$
mrxn

POLAR DECOMPOSITION: For $A \in \mathbb{R}^{n \times n}$, $\exists \underline{Q} \in O(n)$ and $\exists \underline{P} \succeq 0$ such that $A = \underline{Q} \underline{P}$. Note that

$$A = U \Sigma V^T = \underbrace{(UV^T)}_{\substack{!! \\ \underline{Q}}} \underbrace{(V \Sigma V^T)}_{\substack{!! \\ \underline{P}}}$$

Thus: $\min_{V^T X V \succeq 0 \forall X \rightarrow X \succeq 0} \|A - X\|_F = P$

Thm: (ECKART - YOUNG) For $A \in \mathbb{R}^{m \times n}$, write

(Eckart was prof of psychology at UChicago and Young his PhD student)

$$A = U \begin{bmatrix} \sigma_1 & & & & \\ & \ddots & & & \\ & & \sigma_{\min(m,n)} & & \\ & & & & 0 \\ 0 & & & & \vdots \\ & & & & & 0 \end{bmatrix} V^T.$$

Then

$$\min_{\text{rank}(X) \leq r} \|A - X\| = \begin{cases} \sigma_{r+1}, & \text{if } \|\cdot\|_2 \\ \left(\sum_{j=r+1}^{\text{rank } A} \sigma_j^2 \right)^{1/2}, & \text{if } \|\cdot\|_F \end{cases}$$

is attained at

$$X_* := U \begin{bmatrix} \sigma_1 & & & & \\ & \ddots & & & \\ & & \sigma_r & & \\ & & & & 0 \\ 0 & & & & \vdots \\ & & & & & 0 \end{bmatrix} V^T$$

Pf: ($\|\cdot\|_2$) Suppose by contradiction that $\exists B \in \mathbb{R}^{m \times n}$ s.t. $\text{rank } B \leq r$ and $\|A - B\|_2 < \|A - X\|_2 = \sigma_{r+1}$.

Note

$$\|A - X\|_2 = \left\| \begin{bmatrix} 0 & & & & \\ & \ddots & & & \\ & & \sigma_{r+1} & & \\ & & & & 0 \\ 0 & & & & \vdots \\ & & & & & \sigma_{\min(m,n)} \end{bmatrix} \right\|_2 = \sigma_{r+1}.$$

Moreover, $\text{rank } B + \underbrace{\text{nullity } B}_n = n$

$$\dim \ker B \geq n - r$$

Let $\overset{0}{\neq} w \in \ker B$. Then $Bw = 0$ and

$$\|Aw\|_2 = \|(A-B)w\|_2 \leq \|A-B\|_2 \|w\|_2 < \sigma_{r+1} \|w\|_2 - 1.$$

Let $w \in \text{span}\{v_1, \dots, v_{r+1}\} =: W$. Then

$$w = \alpha_1 v_1 + \dots + \alpha_{r+1} v_{r+1} = \begin{bmatrix} v_1 & \dots & v_{r+1} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_{r+1} \end{bmatrix}$$

!!
 $v_{r+1} \alpha$

$$\text{So, } \|Aw\|_2^2 = \|U \Sigma V^T v_{r+1} \alpha\|^2$$

$$= \left\| \Sigma \begin{bmatrix} 1_{r+1} \\ 0 \end{bmatrix} \alpha \right\|^2$$

$$= \left\| \begin{bmatrix} \sigma_1 & \dots & 0 \\ 0 & \dots & \sigma_{r+1} \\ 0 & & 0 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_{r+1} \end{bmatrix} \right\|^2$$

$$= \sum_{j=1}^{r+1} \sigma_j^2 \alpha_j^2$$

$$\geq \sigma_{r+1}^2 \sum_{j=1}^{r+1} \alpha_j^2 = \|w\|^2 \Rightarrow \|Aw\| \geq \sigma_{r+1} \|w\|$$

But $\ker B \cap \text{span}\{v_1, \dots, v_{r+1}\} \neq \{0\}$ ✓

↑
 $\dim = n-r$

↑
 $\dim = r+1$

$> n \Rightarrow$ intersection is non-empty.

MIDTERM: PROVE FOR $\|\cdot\|_F$ →

TOTAL LEAST SQUARES Given $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$

$$(A+E)x = b+r, \text{ where } x, E, r \text{ are unknowns}$$

$\left\{ \begin{array}{l} \text{SOLUTION} \end{array} \right.$

$\uparrow \uparrow$
possible errors from
measurements, etc.

$$\text{Consider } \min_{(A+E)x=b+r} (\|E\|_F^2 + \|r\|_2^2) = \min_{(A+E)x=b+r} \|[E, r]\|_F^2$$

$$\underbrace{Ax - b}_{!!} + \underbrace{Ex - r}_{!!} = 0$$

$$\begin{array}{c} [A, b] \begin{bmatrix} x \\ -1 \end{bmatrix} \\ !! \\ C \in \mathbb{R}^{m \times (n+1)} \end{array}$$

$$\begin{array}{c} [E, r] \begin{bmatrix} x \\ -1 \end{bmatrix} \\ !! \\ F \\ \neq \end{array}$$

$$= \min_{(C+F)z=0} \|F\|_F^2$$

$$\text{and } z := \begin{bmatrix} x \\ -1 \end{bmatrix} \neq 0$$

$$\begin{cases} \text{nullity}(C+F) \geq 1 \\ \text{rank}(C+F) \leq n \end{cases}$$

$$= \min_{\text{rank}(C+F) \leq n} \|F\|_F^2$$

$$C \stackrel{\text{def}}{=} [A, b] \stackrel{\text{SVD}}{=} U \Sigma V^T = U \begin{bmatrix} \sigma_1 & \dots & 0 \\ 0 & \dots & \sigma_{n+1} \\ 0 & & \end{bmatrix} V^T$$

$$= \min_{\text{rank}(C+F) \leq n} \|C - (C+F)\|_F^2$$

$$= \min_{\text{rank}(X) \leq n} \|C - X\|_F^2 \quad \leftarrow \text{APPLY ECKHART-YOUNG}$$

LECTURE 7

Oct 20th, 2025

FLOATING POINT ARITHMETIC

$$F_n := \{2^n : n \in \mathbb{N}\}.$$

Problem: Computer system w/ n bits can only represent 2^n numbers

TYPES OF ERRORS: • ABSOLUTE ERROR: $\|x - x_k\|$ $\leftarrow$ Depends on units

• RELATIVE ERROR: $\frac{\|x - x_k\|}{\|x\|}$

• POINTWISE ERROR: Set $y \in \mathbb{R}^d$, $y_i := \begin{cases} \frac{x_i - x_i^*}{x_i}, & x_i \neq 0 \\ 0, & x_i = 0 \end{cases}$ (Preferred)

IEEE - 754 STANDARD: What #'s to store so that our system is useful.

- Store the sign $\pm$ on the 1st bit

- Always set the decimal place to be after the first #.

$$a = \pm a_1 \cdot 2^0 + a_2 \cdot 2^{-1} + a_3 \cdot 2^{-2} + \dots + a_n \cdot 2^{-n+1}$$

$a_1, \dots, a_n \in \{0, 1\}$ (MANISSA)

$\rightarrow fl: \mathbb{R} \rightarrow F_n$ (floating pt. representation)

THM: (KAHAN) For $x, y \in \mathbb{R}$, $fl(x) = x(1 + \epsilon_0)$, $|\epsilon_0| \leq \epsilon_{\text{machine}}$.

$$fl(x \pm y) = (x \pm y)(1 + \epsilon_1), \quad |\epsilon_1| \leq \epsilon_{\text{machine}};$$

$$fl(x \cdot y) = (x \cdot y)(1 + \epsilon_2), \quad |\epsilon_2| \leq \epsilon_{\text{machine}};$$

$$fl(x/y) = (x/y)(1 + \epsilon_3), \quad |\epsilon_3| \leq \epsilon_{\text{machine}};$$

DEF: $\epsilon_{\text{machine}} := \inf \{x \in \mathbb{R} : x > 0, \text{fl}(1+x) \neq 1\}$.

$$\boxed{\begin{array}{|c|c|c|c|c|c|c|} \hline \pm & a_0 & a_1 & \dots & a_n & e_1 & \dots & e_l \\ \hline \end{array}}$$

$$n = 1 + n + l$$

$$n = 32 \longrightarrow l = 8, n = 23 \quad (\text{SINGLE PRECISION})$$

$$n = 64 \longrightarrow l = 11, n = 52 \quad (\text{DOUBLE PRECISION})$$

SINGLE: 7 decimal digits

$$\epsilon_{\text{mach}} = 2^{-23} \approx 1.2 \cdot 10^{-7}$$

$$10^{-38} \longleftrightarrow 10^{38}$$

$$?? \quad ??$$

$$2^{-128} \longleftrightarrow 2^{128}$$

DOUBLE: 16 decimal digits

$$\epsilon_{\text{machine}} = 2^{-52} \approx 2.2 \cdot 10^{-16}$$

$$10^{-308} \longleftrightarrow 10^{308}$$

$$?? \quad ??$$

$$2^{-1024} \longleftrightarrow 2^{1024}$$

E.g.: Standard precision for telephones/voice $\longrightarrow$ G-117

Deep learning $\longrightarrow$ half-precision (16-bit)

reduced precision (8-bit, 4-bit)

FLOATING POINT ERRORS : $\pm a_1 a_2 a_3 a_4 \cdot 10^e$, $-99 \leq e \leq 99$,
 $a_1, \dots, a_4 \in \{0, 1, \dots, 9\}$

1) ROUNDING ERROR

Storage: $\text{fl}(1.112 \cdot 10^5) = 1.111 \cdot 10^5$

Arithmetic: $\text{fl}(1.111 \cdot 10^4 \cdot 1.111 \cdot 10^2) = 1.234 \cdot 10^3$

Actually: $1.234321 \cdot 10^3$
lost all of these

2) OVERFLOW / UNDERFLOW

When the # is way too large for the FP# system $\leftarrow$ overflow

e.g.: $\text{fl}(1.000 \cdot 10^{55} \cdot 1.000 \cdot 10^{55})$ overflow

When the # is too small for FP# (i.e., too close to zero) $\leftarrow$ underflow

e.g.: Compute $\|x\|_2$.

Don't do $\sqrt{x_1^2 + \dots + x_n^2}$. Instead do: Step 1: $\hat{x} = \begin{cases} x / \|x\|_\infty, & \text{if } \|x\|_\infty \neq 0 \\ 0, & \text{if } \|x\|_\infty = 0 \end{cases}$

Step 2: $\|x\|_2 = \|x\|_\infty \|\hat{x}\|_2$.

3) CANCELLATION ERROR

Subtracting two #'s that are very close together

LECTURE 8

Oct 22nd, 2025

Error: $\|x - \hat{x}\|_{\infty} \leq \varepsilon = 10^{-15} \Rightarrow$ we know that $\hat{x}$ is accurate up to 14 decimal places.
 $\uparrow$ computed quantity
Need to know this error bound

ERROR BOUNDS depend on $\left\{ \begin{array}{l} \text{condition \# of problem} \\ \text{stability of algorithm} \end{array} \right.$

* CONDITIONING OF PROBLEMS

- Consider the linear problem $Ax = b$, $A \in GL(n, \mathbb{R})$

⚠ Actually, because of floating pt. errors $\rightarrow A$ is actually $A + \Delta A$
 b is actually $b + \Delta b$
 $\uparrow$ computed qty.

$$\Rightarrow (A + \Delta A) \hat{x} = b + \Delta b$$

Goal: Bound the relative error $\frac{\|\hat{x} - x\|}{\|x\|} \stackrel{=: \Delta x}{\leq} ?$

PERTURBATION ANALYSIS: Toy example

$$\begin{aligned} A \hat{x} &= b + \Delta b \\ Ax &= b. \end{aligned}$$

$$\Delta x := \hat{x} - x$$

$$\|b\| = \|Ax\| \leq \|A\| \|x\|$$

$$\Rightarrow \frac{1}{\|x\|} \leq \frac{\|A\|}{\|b\|} \quad (2)$$

$$\cancel{A}x + A\Delta x = \cancel{b} + \Delta b$$

$$\Rightarrow A \Delta x = \Delta b$$

$$\Rightarrow \Delta x = A^{-1} \Delta b$$

$$\Rightarrow \|\Delta x\| \leq \|A\|^{-1} \|\Delta b\| \quad (1)$$

$$(1) \cdot (2) \rightarrow \frac{\|\Delta x\|}{\|x\|} \leq \underbrace{\|A\| \|A^{-1}\|}_{\text{CONDITION NUMBER}} \frac{\|\Delta b\|}{\|b\|} \leftarrow \text{relative error in input}$$

relative error in solution

DEF: Condition Number $\kappa_2(A) := \|A\|_2 \|A^{-1}\|_2$

$$\kappa_F(A) := \|A\|_F \|A^{-1}\|_F$$

$$\kappa_*(A) = \|A\|_F \|A^{-1}\|_2, \text{ etc... } p\text{-norms}$$

(HW02)

REMARK: $\frac{\|\Delta x\|}{\|x\|} \leq \frac{\kappa(A) \left[\frac{\|\Delta A\|}{\|A\|} + \frac{\|\Delta b\|}{\|b\|} \right]}{1 - \kappa(A) \frac{\|\Delta A\|}{\|A\|}}$ in general.

$$\frac{1}{1-x} = 1+x+x^2+\dots$$

↑
Only need to know this
up to order of magnitude

ASIDE: SVD implies that every matrix can be written as the product of a rotation, scaling, and another rotation: $A = U \Sigma V^T$.

SPECTRAL CONDITION NUMBER: For $A \in GL(n, \mathbb{R})$

$$\kappa(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} ; \quad \kappa(A) = +\infty \text{ if } A \notin GL(n, \mathbb{R})$$

DEF: $\kappa_2(A) := \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} = \|A\|_2 \|A^+\|_2$

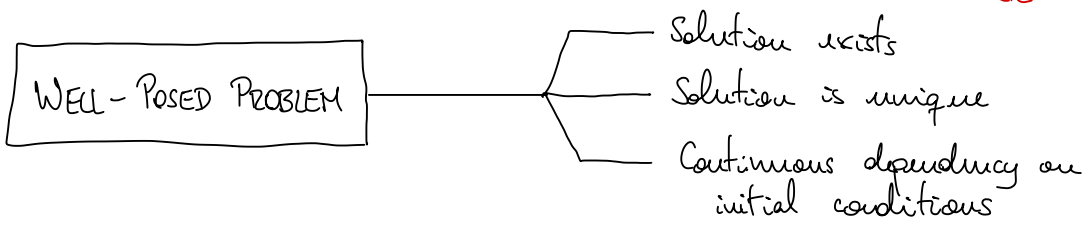
↑
pseudo-inverse

← Always exists for all matrices $A \in \mathbb{R}^{m \times n}$

Similarly, $\kappa_{\text{generalized}}(A) := \|A\| \|A^+\|$ } Exists $\forall$ rectangular matrices $A \in \mathbb{R}^{m \times n}$

REMARK: All of the above condition #'s are for the problem $Ax = b$.

! we have different condition #'s for other problems; e.g., $Ax = \lambda x$, $\min \|Ax - b\|$, $\min C^T x$ s.t. $Ax \leq b$, etc



$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

well-posed

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

ill-posed

Condition number is a continuous measure to how close you are from an ill-posed problem. The bigger the worse!

NOTE: $\kappa(A) = 5 \Rightarrow$ At each operation, we lose $\sqrt{5}$ digits of precision in the answer.

DEF: Given a problem, the condition number of the problem is $\frac{1}{\left(\text{normalized distance to the nearest ill-posed problem}\right)}$.

NOTE: $Ax = b$ $A \in GL(n, \mathbb{R}) \leftarrow$ well-posed region

$A \in M_{\text{ill}} := \{X : \det X = 0\} \leftarrow$ ill-posed region

$$\text{dist}(A, M_{\text{ill}}) = \min_{\det X = 0} \|A - X\|_2 = \min_{\text{rank } X \leq n-1} \|A - X\|_2$$

$$\text{Eckart-Young} \rightarrow \sigma_n(A) \\ = \frac{1}{\sigma_1(A^{-1})}.$$

$$\begin{aligned} \text{Normalize the distance: } \frac{\text{dist}(A, M_{\text{ill}})}{\|A\|_2} &= \frac{1}{\|A\|_2} \cdot \frac{1}{\sigma_1(A^{-1})} \\ &= 1/\kappa_2(A). \quad \checkmark \end{aligned}$$

CONDITION NUMBER AS A DERIVATIVE: Consider

$$\begin{aligned} &\text{(LINEAR SYSTEM)} \quad f: GL(n, \mathbb{R}) \times \mathbb{R}^n \longrightarrow \mathbb{R}^n \\ &\quad (A, b) \longmapsto A^{-1}b = x \end{aligned}$$

$$\begin{aligned} &\text{(MIN. NORM LEAST SQUARES)} \quad f: \mathbb{R}^{m \times n} \times \mathbb{R}^m \longrightarrow \mathbb{R}^n \\ &\quad (A, b) \longmapsto A^+b = x \end{aligned}$$

etc... Upshot: Every problem can be seen as functions (mod measure 0 sets)

DEF: $\kappa_f(A) := \lim_{\delta \rightarrow 0} \sup_{\text{Rel. Error}(a) \leq \delta} \frac{\text{Rel Error}(f(a))}{\text{Rel Error}(a)}$

$$= \lim_{\delta \rightarrow 0} \sup_{\| \Delta a \| < \delta} \frac{\| \Delta f(a) \| / \| f(a) \|}{\| \Delta a \| / \| a \|}$$

$$= \frac{\| Df(a) \|}{\| f(a) \| / \| a \|}.$$

! Easier to compute but assumes problem can be written as a differentiable fct.

EXAMPLE: (CANCELLATION ERROR) Take $f: \mathbb{R}^2 \rightarrow \mathbb{R}$
 $(a, b) \mapsto a - b$

$$Df\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = [1, -1].$$

$$\text{Thus, } \kappa_f\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \frac{\| [1, -1] \|_{\infty}}{|a - b| / \| [a, b] \|_{\infty}} = \frac{1}{|a - b| / \max(a, b)}$$

$$= \frac{\max(a, b)}{|a - b|}$$

$\Rightarrow$ Subtraction is an ill-conditioned problem (Cancellation Error) !

REMARK: $\text{Rel Error}(f(a)) \lesssim \kappa_f(a) \text{ Rel Error}(a)$.

↑
 error bound
 we wanted at
 the beginning

CONDITION NUMBER OF SVD:

$$f: \mathbb{R}^{m \times n} \longrightarrow O(m) \times \mathbb{R}^{\min(m,n)} \times O(n)$$

$$A \longmapsto (U, \Sigma, V)$$

- CRUDE CONDITION # ESTIMATION: Suppose $A = X \Lambda X^{-1}$ is diagonalizable. ↙ WRONG!, but useful...

$$(A + \Delta A) = X(\Lambda + \Delta \Lambda)X^{-1}$$

$$\|\Delta A\| = \|X \Delta \Lambda X^{-1}\| \leq \underbrace{\|X\| \|X^{-1}\|}_{\kappa(X)} \|\Delta \Lambda\|$$

can be arbitrarily large

⇓
Eigenval. problems are, in general, ill-conditioned

✓ Except when...

$$A^T A = A A^T \Rightarrow X = Q \in O(n)$$

$$\Rightarrow \|\Delta A\| \leq \underbrace{\kappa(Q)}_1 \|\Delta \Lambda\|$$

" $\forall Q \in O(n)$

* SVD: $A = U \Sigma V^T \Rightarrow \|\Delta A\| = \|\Delta \Sigma\|$

⇓
SVD can always be computed to machine precision

LECTURE 9

Oct 27th, 2025

* ALGORITHM STABILITY

Model problems as a function $f: X \rightarrow Y$

Approximation b/c of finite-precision arithmetic $\hat{f} \approx f$.

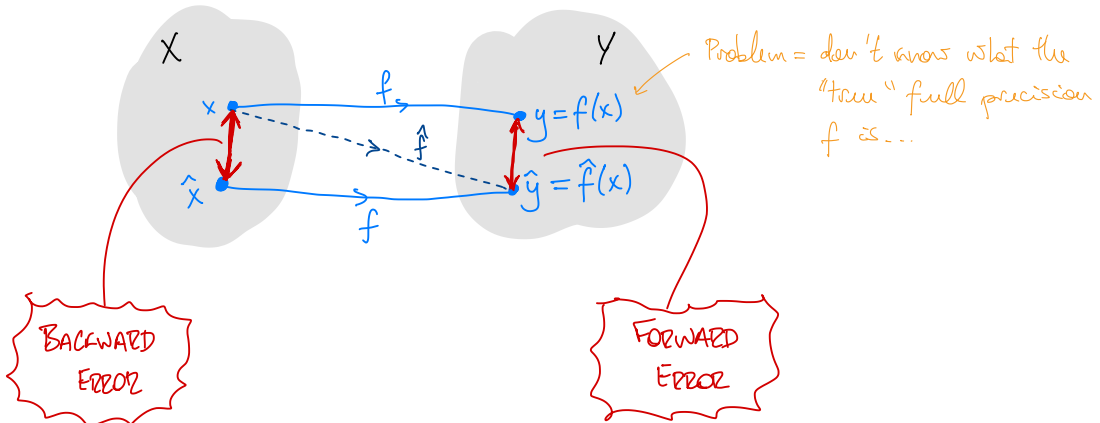
EXAMPLE: (SVD)

$$f: \mathbb{R}^{m \times n} \longrightarrow O(m) \times \mathbb{R}^{m \times n} \times O(n)$$

$$(\hat{U}, \hat{\Sigma}, \hat{V}^T) \stackrel{\hat{f}}{\longleftarrow} A \longmapsto (U, \Sigma, V^T)$$

DEF: • FORWARD ERROR $\rightarrow \|f(x) - \hat{f}(x)\|$

• BACKWARD ERROR $\rightarrow$ Assume $f(\hat{x}) = \hat{f}(x)$ then the backward error is $\|x - \hat{x}\|$



REMARK: $\left(\begin{matrix} \text{FORWARD} \\ \text{ERROR} \end{matrix} \right) \lesssim \left(\begin{matrix} \text{BACKWARD} \\ \text{ERROR} \end{matrix} \right) \cdot \left(\begin{matrix} \text{CONDITION} \\ \text{NUMBER} \end{matrix} \right)$

- For the SVD map above

$$\left(\begin{matrix} \text{forward} \\ \text{error} \end{matrix} \right) = \|\hat{U} - U\| + \|\hat{\Sigma} - \Sigma\| + \|\hat{V} - V^T\|$$

PROBLEM: Don't know what U, Σ, V^T are...

$$\left(\begin{matrix} \text{backward} \\ \text{error} \end{matrix} \right) = \|\hat{U} \hat{\Sigma} \hat{U}^T - A\|$$

Can compute ✓

SMALL BACKWARD ERROR $\equiv$ BACKWARD STABLE

EXAMPLE: (Linear System)

$$f: GL(n, \mathbb{R}) \times \mathbb{R}^n \longrightarrow \mathbb{R}$$

$$(A, b) \longmapsto x := A^{-1}b$$

The approximate version

$$\text{(Gaussian Elimination)} \quad \hat{f}: GL(n, \mathbb{R}) \times \mathbb{R}^n \longrightarrow \mathbb{R}$$

$$(A, b) \longmapsto \hat{x}$$

$$\left(\begin{matrix} \text{forward} \\ \text{error} \end{matrix} \right) = \|\hat{x} - \underbrace{A^{-1}b}_{\text{PROBLEM: We don't know precisely what } A^{-1}b \text{ is...}}\|$$

$$\left(\begin{matrix} \text{backward} \\ \text{error} \end{matrix} \right) = \|Ax - b\|$$

RESIDUAL ✓
(can compute)

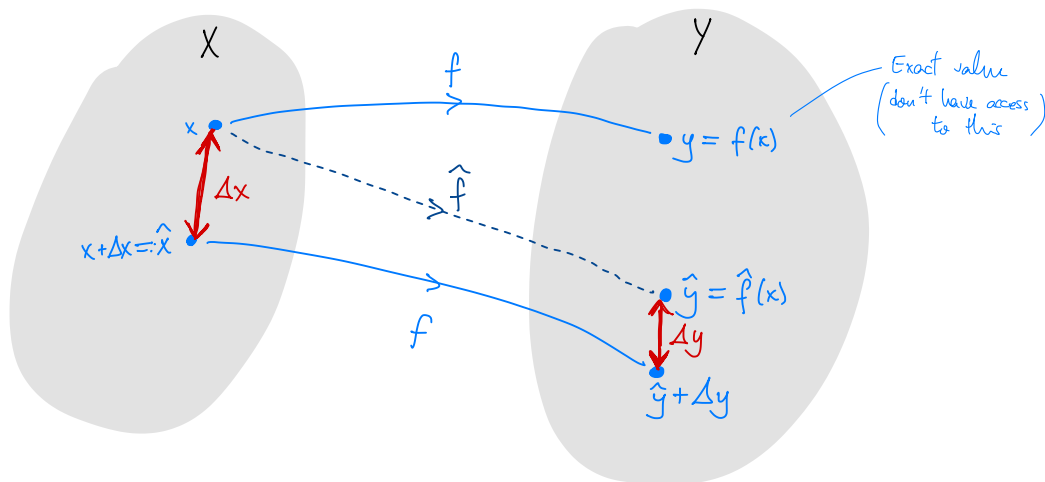
OBS: $\frac{\|\hat{x} - x\|}{\|x\|} \leq \frac{\kappa(A) \frac{\|\Delta A\|}{\|A\|}}{1 - \kappa(A) \frac{\|\Delta A\|}{\|A\|}}$ ← Assume no error in b

$\frac{1}{1-x} \approx x$ ↗

Then: $\frac{\|\Delta x\|}{\|x\|} \approx \underbrace{\kappa(A)}_{\substack{\text{Condition} \\ \text{number}}} \underbrace{\frac{\|\Delta A\|}{\|A\|}}_{\text{Backward error}}$

forward error

LESS NAIVE VERSION (w/out assuming f is onto)



Mixed FORWARD BACKWARD STABILITY $\equiv$ NUMERICAL STABILITY

SMALL $\|\Delta x\|$ and $\|\Delta y\| \Rightarrow \hat{f}$ IS NUMERICALLY STABLE



!

$A = QR \rightarrow R$ is upper triangular
 $\begin{matrix} \cap \\ \mathbb{R}^{m \times n} \end{matrix}$ $\begin{matrix} \cap \\ O(m) \end{matrix}$ $\begin{matrix} \cap \\ \mathbb{R}^{m \times n} \end{matrix}$ $r_{ij} = 0 \quad \forall i > j$

$$= \begin{bmatrix} | & | & \dots & | \\ & & & \end{bmatrix}_{m \times m} \quad \begin{bmatrix} \text{(blue dashed box)} \\ \circ \end{bmatrix}_{m \times n}$$

1) FULL COLUMN-RANK QR (tall-and-skinny matrices)

$$A \in \mathbb{R}^{m \times n}, \quad \text{rank}(A) = n$$

$$A = Q R = Q \begin{bmatrix} R_1 \\ 0 \end{bmatrix}, \quad R_1 \in GL(n, \mathbb{R}) \text{ and } R_1 \text{ is upper triangular}$$

(Gram-Schmidt)

⚠ Absolut nicht erfüllt

 Obs: Need full-col. rank
so that Gram-Schmidt is
well-defined

2) CONDENSED FULL COLUMN-PANK Q12:

$$A = Q R = Q \begin{bmatrix} R_1 \\ \vdots \\ 0 \end{bmatrix}, \quad R_1 \in GL(n, \mathbb{R}) \text{ and } R_1 \text{ is upper triangular}$$

(Gram-Schmidt)

$$= \begin{bmatrix} \overset{n}{Q_1} & \overset{m \times n}{Q_2} \end{bmatrix} \begin{bmatrix} -\frac{P_1}{0} \end{bmatrix} = \underbrace{Q_1 P_1}_{\substack{\text{Condensed QR} \\ \text{factorization}}}$$

3) RANK-RETAINING QR (~~Steiner's Replacement Theorem~~)

Given $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) = r$.

BAD

Permutation Matrix
 $\Pi^{-1} \Pi = I$

$$A \Pi = \begin{bmatrix} A_1 & A_2 \end{bmatrix}$$

permutation matrix
to permute cols.

linearly
independent
columns

linearly
dependent
columns

Better Algorithm:

$$A \Pi = \underbrace{Q}_{O(m)} \begin{bmatrix} R^{r \times r} & S^{r \times (n-r)} \\ 0 & 0 \end{bmatrix}$$



Trash
can

Benefit = gives you the rank by default.

Use this to get (4):

$$\begin{bmatrix} R^T \\ S^T \end{bmatrix} =: Q_z \begin{bmatrix} P_z \\ 0 \end{bmatrix}$$

$$A = \underbrace{Q \begin{bmatrix} R^T & 0 \\ 0 & 0 \end{bmatrix}}_{\text{lower triang.}} \underbrace{Q_z^T P_z^T}_{\text{orthogonal}}$$

4) COMPLETE ORTHOGONAL FACTORIZATION:

Given: $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) = r$

$$A = \underbrace{Q}_{O(m)} \begin{bmatrix} L^{r \times r} & 0 \\ 0 & 0 \end{bmatrix} \underbrace{Z^T}_{O(n)}$$

$L \in \mathbb{R}^{r \times r}$ lower-triangular ($L \in GL(r, \mathbb{R})$)

Obs: Can be done in finitely-many steps (unlike SVD, EVD, schur, etc)

LECTURE 10

QR - DECOMPOSITION

Oct 29, 2025

QR DECOMPOSITION

! WLOG, assume square matrix b/c we can append cols/rows to make rect $\rightarrow$ square

- Given $A \in \mathbb{R}^{n \times n}$, A full-rank, $A = QR$.

$$[a_1 \dots a_n] = A = [q_1 \dots q_n] \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ & r_{22} & & \vdots \\ & & \ddots & \\ 0 & & & r_{nn} \end{bmatrix}$$

Thus:

$$\begin{cases} a_1 = r_{11} q_1 \\ a_2 = r_{12} q_1 + r_{22} q_2 \\ \vdots \\ a_n = r_{1n} q_1 + \dots + r_{nn} q_n \end{cases}$$

Have QR decomp.
 $\Updownarrow$
Solution to system

Note: (i) From the 1st eq. in the system

$$\|a_1\| = r_{11} \|q_1\| = r_{11}$$

$\leftarrow$ no problem here with a_1 being zero b/c we assumed A is full-rank.

$$\Rightarrow a_1 = \|a_1\| q_1, \text{ i.e., } q_1 = \frac{a_1}{\|a_1\|}$$

$$(ii) a_2^T q_1 = r_{12} q_1^T q_1 + r_{22} q_2^T q_1 \Rightarrow r_{12} = a_2^T q_1$$

$\Rightarrow 0$ b/c $Q \in O(n)$

$$\text{So, } r_{22} \|q_2\| = \|a_2 - (a_2^T q_1) q_1\| \Rightarrow r_{22} = \|a_2 - (a_2^T q_1) q_1\|$$

$$\text{Thus: } q_2 = \frac{a_2 - (a_2^T q_1) q_1}{\|a_2 - (a_2^T q_1) q_1\|}$$

(iii) Repeat

$$(k) \quad a_k = \sum_{j=1}^k r_{jk} q_j \quad \begin{cases} \rightarrow r_{jk} = q_j^T a_k, j=1, \dots, k \\ \rightarrow q_k = \frac{1}{r_{kk}} \left[a_k - \sum_{j=1}^{k-1} r_{jk} q_j \right] \end{cases}$$

$$r_{kk} = \left\| a_k - \sum_{j=1}^{k-1} r_{jk} q_j \right\| \neq 0 \text{ b/c } A \text{ is full-rank}$$

NOTE: We made the choice to make $\{r_{kk}\}_{k=1}^n$ all positive.
 But could have chosen them to be negative too...
 → Useful to make some algorithms more stable

! IF A is NOT FULL-RANK: $\text{rank } A = r < n$

KEY: Permute the linearly dependent cols. to the end

$$A = [a_1 \dots a_n] \xrightarrow{\text{permute}} A\Pi = [A_1, A_2] = Q \begin{bmatrix} R & S \\ 0 & 0 \end{bmatrix}$$

STUDY FOR MIDTERM: HOW TO GET S !

REMARK: Gram-Schmidt is UNSTABLE

Orthogonal or unitary

$$A = U \Sigma V^T, \quad A = Q \Lambda Q^T, \quad A = QR$$

$$A = Q \begin{bmatrix} L & 0 \\ 0 & 0 \end{bmatrix} Z^T, \quad A = LZ, \quad A = Z^T Z$$

Types of decomposition blocks:

- Diagonal: Σ, Λ
- Triangular: L, R
- Orthogonal (or unitary...): U, V, Q, Z
- Structured matrices, tridiagonals, Toeplitz, etc...

* BACK SUBSTITUTION ← **STUDY PROBLEM ON THIS FOR TEST**

Note:

$$\begin{bmatrix} r_{11} & & & \\ & r_{22} & & \\ & & \ddots & \\ & & & r_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} \quad \begin{matrix} \text{Common} \\ \text{sense} \end{matrix} \quad x_n = \frac{b_n}{r_{nn}}.$$

Iterate as back substitution

$$x_{n-1} = \frac{b_{n-1} - r_{nn}x_n}{r_{n-1,n-1}}, \quad \dots, \quad x_1 = \dots$$

DEF: Let $A \in \mathbb{R}^{m \times n}$ with $\text{rank } A = r$. We can decompose A as

$$A = G H, \quad G \in \mathbb{R}^{m \times r}, \quad H \in \mathbb{R}^{r \times n}.$$

This is called RANK-RETAINING DECOMPOSITION

→ **WILL APPEAR ON MIDTERM**
PROVE THIS!

LEMMA: (1) $G^T G$ is invertible

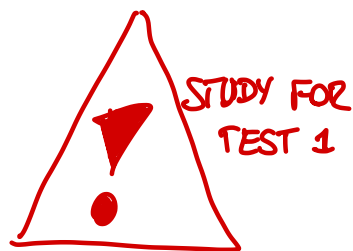
(2) $H H^T$ is invertible

(3) $\text{im}(A) = \text{im}(G)$

(4) $\ker(A^T) = \ker(G^T)$

(5) $\ker(A) = \ker(H)$

(6) $\text{im}(A^T) = \text{im}(H^T)$



EXAMPLE: $A = U \Sigma V^T \xrightarrow[\text{SVD}]{\text{condensed}} A = \underbrace{U_r}_{G} \underbrace{\Sigma_r V_r^T}_{H}$

$$\begin{bmatrix} U_r & U_{m-r} \end{bmatrix} \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_r \\ V_{n-r} \end{bmatrix}$$

REMARK: Can solve least norm least squares with this:

$$x^+ \in \operatorname{argmin} \{ \|x\| : x \in \operatorname{argmin} \|Ax - b\| \},$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$. Write $\begin{cases} \mathbb{R}^n = \ker A \oplus \operatorname{im} A^T \\ \mathbb{R}^m = \operatorname{im} A \oplus \ker A^T \end{cases}$

So, $b = \underset{\substack{\uparrow \\ \mathbb{R}^m}}{b_0} + \underset{\substack{\uparrow \\ \ker A^T}}{b_1}$, moreover $b_0 \perp b_1$ (Fredholm's Alternative).
Thus, $b_0 \perp b_1$

$$\|Ax - b\|^2 = \|Ax - b_0 - b_1\|^2 = \|Ax - b_1\|^2 + \|b_0\|^2.$$

So, the $x \in \mathbb{R}^n$ that solves this is the x that $x \xrightarrow{A} b$. So,
 $Ax - b_1 = 0$.

Use rank-preserving decomposition: $A = GH$. Then, by the lemma above (PROVE IT!) $\operatorname{im} A = \operatorname{im} G$, hence

$$Ax = b_1 \rightsquigarrow GHx = Gs \text{ for some } s \in \mathbb{R}^r \quad \text{b/c } b_0 \in \ker A^T = \ker G^T$$

$$G^T GHx = G^T Gs = G^T b_1 = G^T b_1 + \cancel{G^T b_0} = G^T b_1$$

$$\text{blue } G^T G \text{ invertible} \iff Hx = (G^T G)^{-1} G^T b_0$$

Decompose $x = \underset{\substack{\uparrow \\ \ker A}}{x_0} + \underset{\substack{\uparrow \\ \operatorname{im} A^T}}{x_1}$, $x_0 \perp x_1$, then $\|x\|^2 = \|x_0\|^2 + \|x_1\|^2$.
 $\operatorname{im} A^T = \operatorname{im} H^T \Rightarrow x_1 = H^T y \text{ for some } y$

$$\text{i.e., } Hx = \cancel{Hx_0} + Hx_1 = Hx_1 \iff Hx_1 = Hx = (G^T G)^{-1} G^T b_0$$

$$\Leftrightarrow x_1 = H^T (H H^T)^{-1} (G^T G)^{-1} G^T b_0$$

$$\begin{aligned} & \parallel \\ & x^+ \in \operatorname{argmin} \{ \|x\| : x \in \operatorname{argmin} \|Ax - b\| \} \\ & \parallel \\ & A^+ b \end{aligned}$$

MEMORIZE $\downarrow$

Upshot: We can write the pseudo-inverse as:

$$A^+ = H^T (H H^T)^{-1} (G^T G)^{-1} G^T$$



_____ // _____

* FULL-RANK LEAST SQUARES

• Given: $A \in \mathbb{R}^{m \times n}$, $\operatorname{rank} A = n$ $\leftarrow$ more cols. than rows

• WTF: $\min \|Ax - b\|$

NOTE: $A = QR$ $\downarrow$

$$\begin{aligned} \|Ax - b\|^2 &= \left\| Q \begin{bmatrix} R_1 \\ 0 \end{bmatrix} x - b \right\|^2 \\ &= \left\| \begin{bmatrix} R_1 \\ 0 \end{bmatrix} x - Q^T b \right\|^2 \quad \text{where } Q^T b = \begin{bmatrix} c \\ d \end{bmatrix} \\ &= \left\| \begin{bmatrix} R_1 x \\ 0 \end{bmatrix} - \begin{bmatrix} c \\ d \end{bmatrix} \right\|^2 \end{aligned}$$

$$= \left\| \begin{bmatrix} R_1 x - c \\ -d \end{bmatrix} \right\|^2$$

$$= \|R_1 x - c\|^2 + \|d\|^2$$

Back Substitution $\rightarrow R_1 x = c$

Upshot: Solver to the full-rank least squares QR METHOD

1. Form QR decomposition $A = QR$

2. Set $Q^T b = \begin{bmatrix} c \\ d \end{bmatrix}$

3. $R_1 x = c$ solve through back substitution

Output x .

(Very
stable
numerically)

- NORMAL EQUATION METHOD (fronned upon in numerical analysis)
(to solve full-rank least squares)

$Ax = b$ rewrite $\rightarrow$ $A^T A x = A^T b$ Always has a solution



WARNING: This is ILL-CONDITIONED ($\kappa(A^T A) \approx \kappa(A)^2$)
and UNSTABLE

cannot store this...
 $fl(1+\delta^2) = 1$

$$A = \begin{bmatrix} 1 & 1 \\ \delta & 0 \end{bmatrix} \rightarrow A^T A = \begin{bmatrix} 1+\delta^2 & 1 \\ 1 & 1 \end{bmatrix}$$

end up with $A^T A$ singular

REMARK: Usually $m \gg n$.

- Condition # of QR METHOD:

$$\frac{\|\hat{x} - x\|}{\|x\|} \leq 2\kappa_{QR}(A)\varepsilon + \kappa_{QR}(A)^2 \frac{\|b - Ax\|_2}{\|A\| \|x\|}$$

- Condition # of NORMAL EQUATION METHOD:

$$\frac{\|\hat{x} - x\|}{\|x\|} \leq \kappa_{NE}(A)^2 \left(1 + \frac{\|b\|}{\|A\| \|x\|} \right) \varepsilon$$

REMARK: Normal Equation is better when $m \gg n$ (e.g. $n=3$ and $m=100,000$) b/c then we only have to solve a 3×3 system. Actually, better than forming the QR-decomposition.

• AUGMENTED SYSTEM METHOD:

$$A^T A x = A^T b$$

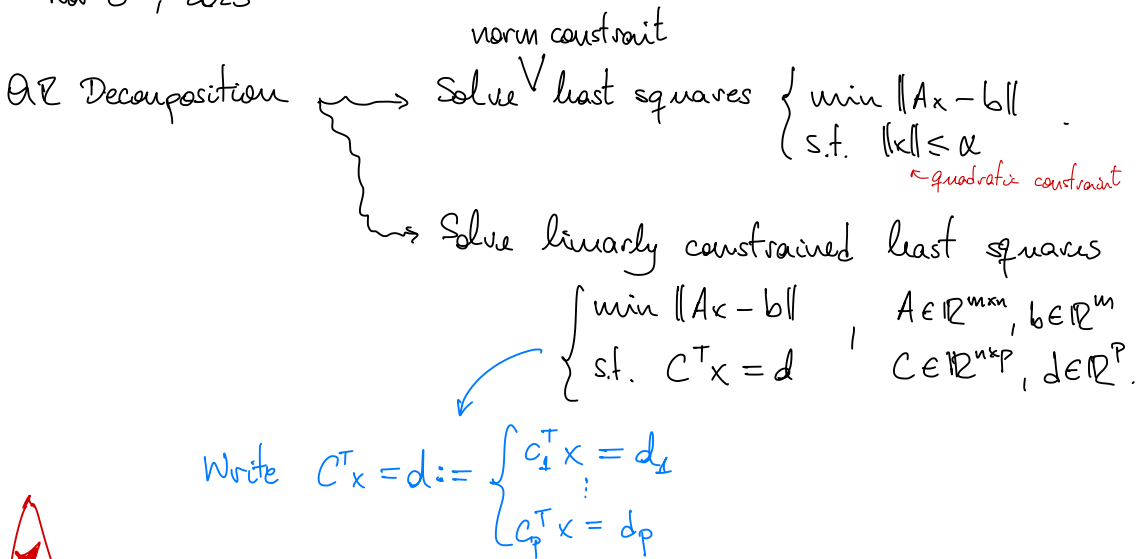
→ keep the nice structures (sparsity, tridiagonality, etc) of A .

$$\begin{cases} r := Ax - b \\ A^T r = 0 \end{cases} \rightsquigarrow \begin{bmatrix} I & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} r \\ x \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix} \quad (\text{AUGMENTED SYSTEM})$$

PROBLEM: Increased dimensions by a lot

LECTURE 11

Nov 3rd, 2025



LAGRANGE MULTIPLIERS : $\mathcal{L}(x, \underbrace{\lambda_1, \dots, \lambda_p}_{\lambda}) := \|Ax - b\|^2 + 2 \sum_{i=1}^p \lambda_i (c_i^T x - d_i)$

!!

$$\mathcal{L}(x, \lambda) = \|Ax - b\|^2 + 2 \lambda^T (C^T x + d)$$

1) Differentiate w.r.t. x and set to zero:

$$0 \stackrel{!}{=} \nabla_x \mathcal{L}(x, \lambda) = 2(A^T A x - A^T b + C \lambda)$$

2) Differentiate w.r.t. λ and set to zero $\leftarrow$ Don't waste time b/c this only gives back the constraints

$$0 \stackrel{!}{=} \nabla_\lambda \mathcal{L}(x, \lambda) = C^T x = 0$$

- METHOD 1: Rewrite as matrix $\begin{bmatrix} A^T A & C \\ C^T & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} A^T b \\ 0 \end{bmatrix}$.

Problem: $\kappa(A^T A) \approx \kappa(A)^2$. Good: Analytically

- METHOD 2:
- 1) $\hat{x} \in \operatorname{argmin} \|Ax - b\|$
 - 2) $C^T(A^T A)^{-1} C \lambda = C^T x - d$ to get λ_*
 - 3) $A^T A x = A^T b - C \lambda_*$ $\rightarrow x = \underbrace{(A^T A)^{-1} A^T b}_{\hat{x}} - \underbrace{(A^T A)^{-1} C \lambda_*)_m$

This works because of METHOD 1 and $A^T A x = A^T b$ bad conditioning... ↙
METHOD 2 can be implemented without forming any $M^T M$
and requires only two QR's:

ALGORITHM FOR METHOD 2 (Suppose A is full-rank)

- INPUT: A, b, C, d

- STEP 1: Compute $A = Q \begin{bmatrix} R \\ 0 \end{bmatrix}$, $R \in GL(n, \mathbb{R})$
- STEP 2: Solve $\min \|Ax - b\|$ to get $\hat{x}$ by using QR from Step 1.
- STEP 3: Form $W = R^{-T} C$ by doing p back substitutions to solve the following linear systems: good for triangular systems
 $R^T w_i = c_i, \quad i = 1, \dots, p.$
- STEP 4: Compute $W = Q_\perp R_\perp$
- STEP 5: Form $\eta := C^T x - d$

- STEP 6: Solve $R_1^T R_1 \lambda = \eta$ using two back substitutions
as follows: good for triangular systems

$$\begin{cases} R_1^T u = \eta \\ R_1 \lambda = u \end{cases}$$

- STEP 7: Set $x = \hat{x} - \underbrace{(R^T R)^{-1} C \lambda}_{=: \xi}$ but don't actually compute the inverse. Compute ξ using back substitutions in the following triangular systems

$$\begin{cases} R^T \xi = C \lambda \\ R \xi = \xi \end{cases}$$

$\uparrow$ dummy variable

— OUTPUT: x .

REMARK: Solving $\begin{cases} \min \|Ax - b\| \\ \text{s.t. } C^T x = b \end{cases}$ involves n variables of x and

the constraint should decrease the # of possible solutions.

But, solving the system from METHODS 1 & 2 actually increase the dimensionality of the problem:

$$\begin{bmatrix} A^T A & C \\ C^T & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} A^T b \\ d \end{bmatrix}.$$

$n+p$ variables...

(METHOD 3 below fixes this)

- METHOD 3:

1) Compute the QR of the feasible set

$$C := Q_2 \begin{bmatrix} Z_2 \\ 0 \end{bmatrix} \begin{matrix} p \\ n-p \end{matrix}$$

$n \times p$ $n \times n$

2) Then, the constraint takes the analytic form

$$\begin{cases} Z_2^T u = d \\ Q_2^T x = \begin{bmatrix} u \\ v \end{bmatrix} \end{cases}$$

$v \in \mathbb{R}^{n-p}$

3) Consider $\|b - Ax\| = \|b - \underbrace{A Q_2}_{\substack{=: \tilde{A} \\ \tilde{A} = A Q_2}} \underbrace{Q_2^T x}_{\substack{= \begin{bmatrix} u \\ v \end{bmatrix}}} \| = \|b - \tilde{A} \begin{bmatrix} u \\ v \end{bmatrix}\|$

$$= \|b - \tilde{A} \begin{bmatrix} u \\ v \end{bmatrix}\|$$
$$= \|b - [\tilde{A}_1 \ \tilde{A}_2] \begin{bmatrix} u \\ v \end{bmatrix}\|$$
$$= \|b - \tilde{A}_1 u - \tilde{A}_2 v\|.$$

ALGORITHM FOR METHOD 3

- INPUT: A, b, C, d

• STEP 1: Compute $C = Q_2 Z_2$

• STEP 2: Compose $\tilde{A} = A Q_2$

- STEP 3: Solve $\tilde{R}_2^T u = d$ using back substitution to get $\hat{u}$
- STEP 4: Solve $\hat{v} \in \arg\min \| (b - \tilde{A}_1 \hat{u}) - \tilde{A}_2 v \|$ ← unconstrained least squares
- STEP 5: Set $x := \Theta \begin{bmatrix} \hat{u} \\ \hat{v} \end{bmatrix}$.
- OUTPUT: x .

REMARK: METHOD 3 has less unknowns. Also,

$$\kappa_2(\tilde{A}_2) \leq \kappa_2(\tilde{A}) = \kappa(A).$$

κ_2 measures "how linearly independent a set of vectors are"

———— // ————

COMPUTING QR DECOMPOSITION

1) GRAM-SCHMIDT: $A R_1^{-1} R_2^{-1} \cdots R_{n-1}^{-1} = Q$

$$A = Q \underbrace{R_{n-1} R_{n-2} \cdots R_1}_{=R} = QR$$

BAD: Condition # of upper triangular can be arbitrarily large

⇒ Gram-Schmidt is very numerically unstable

2) Do the opposite: find an upper-triangular using orthogonal matrices

$$Q_{n-1}^T \cdots Q_2^T Q_1^T A = R$$

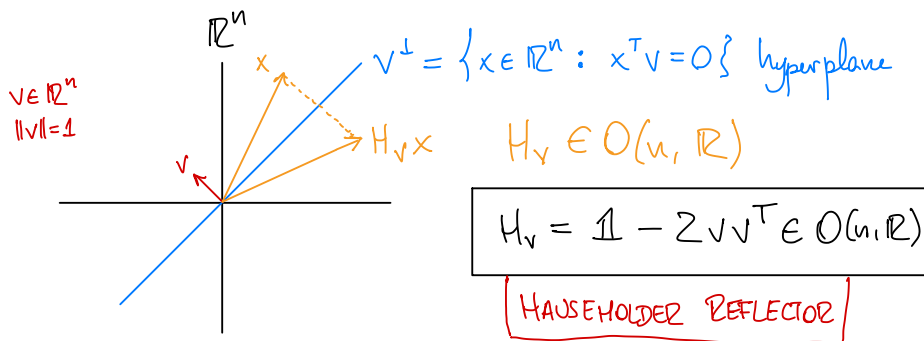
$$A = \underbrace{Q_1 Q_2 \cdots Q_{n-1}}_{\substack{=: \\ Q}} R = QR.$$

⇒ Well-conditioned because condition # of orthogonal matrices is always $\equiv 1$.



→ [Alton Householder got his Math PhD at UChicago ☺]

* HOUSEHOLDER REFLECTORS



NOTE: (i) $H_v H_v = H_v^2 = \mathbb{1}$ (H_v is an INVOLUTION)

(ii) $H_v^T H_v = \mathbb{1} = H_v H_v^T$ ($H_v \in O(n, \mathbb{R})$)

REMARK: If $v \in \mathbb{R}^n$ not normalized, take $H_v = \mathbb{1} - \frac{vv^T}{v^T v}$.

STEP 1: Want to eliminate the 1st column by

$$Q^T A = \begin{bmatrix} \delta & \boxed{*} \\ \vdots & \\ 0 & \end{bmatrix} \rightsquigarrow Q^T \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \stackrel{!}{=} \begin{bmatrix} \delta \\ 0 \\ \vdots \\ 0 \end{bmatrix} = r e_1$$

[pick the "+" one here, but can also pick "-" if it's better.] $\rightsquigarrow r = \pm \sqrt{a_1^2 + \dots + a_n^2} = \|a\|$

STEP 2: Take

$$H_v a \stackrel{\text{def}}{=} (I - 2vv^T)a = \pm \|a\| e_1.$$

Since H_v is an involution,

$$a = H_v^2 a = \pm \|a\| (e_1 - 2\underbrace{vv^T e_1}_{=v_1})$$

i.e.,
$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \underbrace{\pm \|a\|}_{\substack{!! \\ \alpha}} \begin{bmatrix} 1 - 2v_1^2 \\ -2v_1 v_2 \\ \vdots \\ -2v_1 v_n \end{bmatrix}$$

$$\Rightarrow v_1 = \pm \sqrt{\frac{1}{2} \left(1 - \frac{a_1}{\alpha}\right)} \quad ; \quad v_j = -\frac{a_j}{2\alpha v_1}, \quad j=2, \dots, n.$$

STEP 3: So far

$$H_1^T A = \begin{bmatrix} \alpha & \boxed{*} & * & * & * & * \\ 0 & \boxed{*} & * & * & * & * \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \boxed{*} & * & * & * & * \end{bmatrix}$$

as desired.

RESULT

$$H_{n-1}^T H_{n-2}^T \dots H_1^T A = R$$

$\hat{A}_2$ and take $\hat{A}_2$ and apply steps 1&2 above to $\hat{A}_2$

α_2 $\begin{bmatrix} * & * & * & * & * \\ 0 & * & * & * & * \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & * & * & * & * \end{bmatrix}$ $\rightarrow$ take this as $\hat{A}_3$ and repeat

LECTURE 12

Nov 10th, 2025

GIVENS ROTATIONS & QR DECOMPOSITION

WTF: An orthogonal matrix that gives us QR

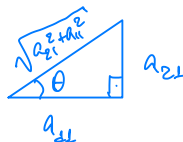
$$\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} r & * \\ 0 & * \end{bmatrix} \quad \left. \begin{array}{l} \gamma := \cos\theta \\ \sigma := \sin\theta \end{array} \right\} \Rightarrow \gamma^2 + \sigma^2 = 1$$

$$\begin{bmatrix} \gamma & \sigma \\ -\sigma & \gamma \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix}$$

$$\Rightarrow -\sigma a_{11} + \gamma a_{21} = 0 \iff \gamma^2 a_{21}^2 = \sigma^2 a_{11}^2$$

$$= (1 - \gamma^2) a_{11}^2$$

$$\Rightarrow \gamma = \pm \frac{a_{11}}{\sqrt{a_{21}^2 + a_{11}^2}}$$



$$\text{and so } \sigma = \pm \frac{a_{21}}{\sqrt{a_{21}^2 + a_{11}^2}}$$

Moreover, $r_{11} = \pm \sqrt{a_{21}^2 + a_{11}^2}$

GENERALIZE:

$$\begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & \boxed{\gamma} & \boxed{\sigma} \\ & & \boxed{-\sigma} & \boxed{\gamma} \\ & & & \ddots & 1 \end{bmatrix} \begin{bmatrix} * \\ \vdots \\ \alpha \\ \beta \\ * \end{bmatrix} = \begin{bmatrix} * \\ \vdots \\ \sqrt{\alpha^2 + \beta^2} \\ * \\ * \end{bmatrix}$$

Other entries are just zero

Labels in diagram: i^{th} and j^{th} rows and columns are indicated for the rotation submatrix. The rotation is applied to the i^{th} and j^{th} rows of the matrix.

$$G_{ij}(\theta)$$

Doing this, we can introduce zeros anywhere we need on the vector on the right

REMARK: Don't really care about θ itself, as long as we get the zeros on the right.

REMARK: To do QR on an $n \times n$ matrix, we need $O(n^2)$ Givens rotations (cf $O(n-1)$ from Householder reflectors).

REMARK: Very useful for matrices that are almost in upper triangular, for ex.:

$$\begin{array}{l} \sqrt{\alpha^2 + \beta^2} \rightarrow \begin{bmatrix} * & * & \dots & * \\ & * & \dots & * \\ & & 0 & \vdots \\ * & * & \dots & * \end{bmatrix} \\ \text{becomes } 0 \rightarrow \begin{array}{l} \text{becomes zero} \quad \text{becomes zero} \end{array} \end{array}$$

(SEE NOTES) !

COMPLETE ORTHOGONAL FACTORIZATION: Combine Householder with Givens to enforce linear independence!

$$\underbrace{H_{n-1} \dots H_1}_{Q^T} \begin{bmatrix} A_1 & A_2 & \dots & A_n \\ * & * & \dots & * \\ * & * & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \dots & * \end{bmatrix} \underbrace{\Pi_1 \dots \Pi_{n-1}}_{= \Pi} = \begin{bmatrix} R & S \\ 0 & 0 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{= A}$

$\|A_i\| > \dots > \|A_j\| = \|A_r\| = \dots$

Now, redo the above with

$$\hat{H}_{r-1} \cdots \hat{H}_1 \begin{bmatrix} R^T \\ S^T \end{bmatrix} = \begin{bmatrix} L^T \\ 0 \end{bmatrix} \Rightarrow A = Q \begin{bmatrix} L & 0 \\ 0 & 0 \end{bmatrix} Z^T,$$

$$Z = \Pi Q$$

GAUSSIAN ELIMINATION (GE)

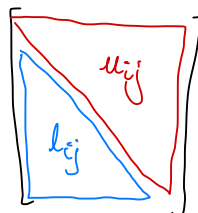
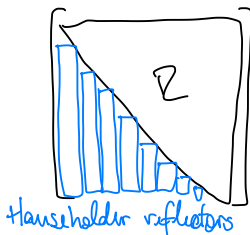
← Way to get LU factorization

LU FACTORIZATION:

$$\mathbb{R}^{n \times n} \ni A = LU, \quad U = \begin{bmatrix} u_{11} & \cdots & u_{1n} \\ & \ddots & \vdots \\ 0 & & u_{nn} \end{bmatrix} \leftarrow (\text{upper triangular})$$

$$L = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ (l_{ij}) & & 1 \end{bmatrix} \leftarrow \begin{array}{l} \text{Row-Echelon} \\ \text{operations to} \\ \text{get reduced form} \\ \text{(unit lower triangular)} \end{array}$$

REMARK: Can store QR and LU decompositions without having to use more memory by



Already know the diagonal of L is just 1's.

Note: $H_v = \mathbb{1} - 2vv^T$ (rank-1 perturbation of the $\mathbb{1}$)

$$M_i = \mathbb{1} - m e_i^T, \quad m \in \mathbb{R}^n \quad (\text{Gaussian elimination Matrix})$$

$$M_1 a = \begin{bmatrix} \gamma \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \gamma e_1 \leadsto (\mathbb{1} - m e_1^T) a = \gamma e_1$$

So: $(\mathbb{1} - m_1 e_1^T) a = \gamma e_1, \quad a = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix},$

$$\parallel$$

$$a - e_1^T a m_1$$

$$\parallel$$

$$a_1 - a_1 m_1$$

where $a_1 \neq 0$

big assumption

can be big/small b/c M is not orthogonal (unlike H)

Gaussian elimination is unstable.

If $a_1 \neq 0$, $m_1 = \frac{1}{a_1} (a - \gamma e_1) = \begin{bmatrix} 0 \\ a_2/a_1 \\ \vdots \\ a_n/a_1 \end{bmatrix}.$

Choose $\gamma = a_1$

$$M_1 = \begin{pmatrix} 1 & & & 0 \\ -a_2/a_1 & 1 & & \\ \vdots & & \ddots & \\ -a_n/a_1 & 0 & & 1 \end{pmatrix}.$$

$-l_{i1} = -\frac{a_i}{a_1}$

Ex: $A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}, \quad M_1 A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22}^{(2)} & \dots & a_{2n}^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{m2}^{(2)} & \dots & a_{mn} \end{bmatrix}$

$(a_{11} \neq 0)$

$A^{(2)} \in \mathbb{R}^{(m-1) \times (n-1)}$

$$M_2 M_1 A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & & & \\ \vdots & & & \\ 0 & & & \end{bmatrix}$$

$A^{(2)}$

$$M_2 = \mathbb{I} - m_2 e_2^T, \quad m_2 = \begin{bmatrix} 0 \\ 1 \\ -a_{32}^{(2)} / a_{22}^{(2)} \\ \vdots \\ -a_{m2}^{(2)} / a_{22}^{(2)} \end{bmatrix} =: \begin{bmatrix} 0 \\ 1 \\ -l_{32} \\ \vdots \\ -l_{m2} \end{bmatrix},$$

$$l_{i2} = \frac{a_{i2}^{(2)}}{a_{22}^{(2)}}.$$

⋮ Iterate

$$a_{kk}^{(k)} \neq 0, \quad k=1, 2, \dots, m-1$$

$$M_k = \begin{pmatrix} 1 & & 0 & & \\ & \ddots & & & \\ & & 1 & & \\ 0 & & -l_{k+1,k} & 1 & 0 \\ & & \vdots & & \ddots \\ & & -l_{m,k} & & 1 \end{pmatrix}, \quad l_{ik} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$$

$$\underbrace{M_{m-1} M_{m-2} \dots M_1}_{\substack{!! \\ L^{-1}}} A = \begin{bmatrix} \text{upper triangular} \\ 0 \end{bmatrix} =: U$$

$$\Rightarrow L^{-1} = M_1^{-1} \dots M_{m-2}^{-1} M_{m-1}^{-1}$$

NOTE: M_k^{-1} is easily computed

b/c

$$M_k = \mathbb{I} - m_k e_k^T$$

$$\} \\ M_k^{-1} = \mathbb{I} + m_k e_k^T$$

$$\Leftrightarrow M_k^{-1} = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & l_{k+1,k} & 1 & 0 \\ & & \vdots & & \ddots \\ & & l_{m,k} & & 1 \end{bmatrix}$$

DEF: Matrices with $a_{kk}^{(k)} \neq 0$ for all $k=1, \dots, n$ is called
STRONGLY NONSINGULAR MATRIX. $\leftarrow$ dense set in the space of matrices

$$\Leftrightarrow \det A^{(k)} \neq 0, \quad A^{(k)} \text{ } k\text{-minor}$$

$$\Leftrightarrow A = LU \text{ with } L \text{ unit lower triangular and } U \text{ upper triangular}$$

CLAIM: A sufficient condition for strongly nonsingular:



A is strictly diagonally dominant

CHECK

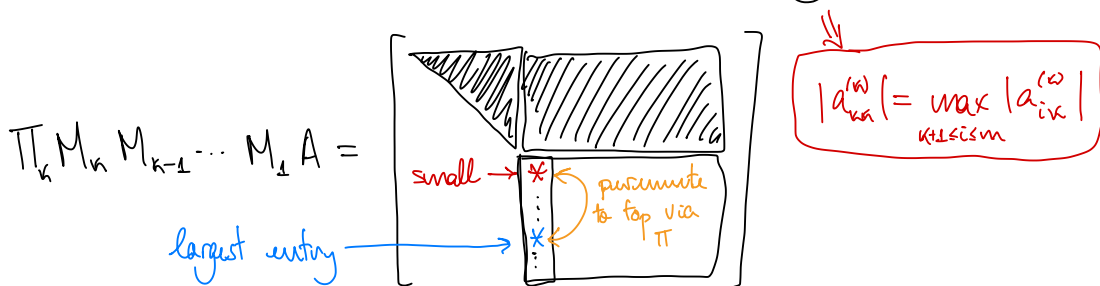


A is strongly nonsingular



WARNING: Dividing by $a_{kk}^{(k)}$ very small is BAD (fl.pt. arithmetic)
 $\Rightarrow$ Pivoting

PIVOTING: Partial Pivoting $\rightarrow$ Gaussian elimination with
Partial Pivoting = GEPP.



ACTUAL LU: Partial Pivoting

$$M_m \Pi_m M_{m-1} \Pi_{m-1} \dots M_2 \Pi_2 M_1 \Pi_1 A = \begin{bmatrix} \text{shaded triangle} \\ 0 \end{bmatrix} = U$$

READ NOTES

$\Rightarrow \Pi A = LU$, $\Pi = I$ only for strongly nonsingular matrices A

LECTURE 13

Nov 12, 2025

LU WITH PARTIAL PIVOTING: (GEPP) $M_k := I - m_k e_k^T$

$$A \rightarrow \dots \rightarrow \begin{bmatrix} \text{shaded triangle} & \text{shaded rectangle} \\ \text{red lines} & A_{kk}^{(k)} \end{bmatrix} \rightarrow \dots \rightarrow \begin{bmatrix} U \\ L \end{bmatrix}$$

(l_{ij})

$$M_2 \Pi_2 M_1 \Pi_1 A = U$$

$$M_1 = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}$$

$$\tilde{M}_1 = \begin{bmatrix} 1 & & \\ & 1 & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

$$M_2 \underbrace{(\Pi_2 M_1 \Pi_1^T)}_{\tilde{M}_1} \underbrace{\Pi_2 \Pi_1}_{\Pi} A = U$$

$$M_2 \tilde{M}_1 \Pi A = U$$

$$\underbrace{M_2 \tilde{M}_1}_{L^{-1}}$$

$$L^{-1} \Pi A = U \longrightarrow \boxed{\Pi A = LU}$$

• NOTE:

$$L = \begin{bmatrix} 1 & & & 0 \\ & \ddots & & \\ (l_{ij}) & & \ddots & \\ & & & 1 \end{bmatrix}, \quad |l_{ik}| = \left| \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} \right| \leq 1$$

$$\Rightarrow |l_{ij}| \leq 1 \quad \forall i, j.$$

• NOTE: $|a_{kk}^{(k)}| = \max_{k \leq i \leq n} |a_{ik}^{(k)}| \leftarrow \text{From pivoting}$

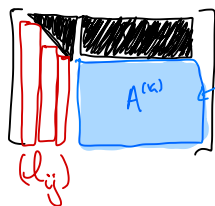
(GROWTH FACTOR) $\gamma_{\text{GEPP}} := \max_{\substack{1 \leq i \leq n \\ 1 \leq j \leq n \\ 1 \leq k \leq n}} \frac{|a_{ij}^{(k)}|}{|a_{ij}|} \left. \begin{array}{l} \text{Can be as large as } 2^{n-1} \text{ but,} \\ \text{in practice, this seems hardly} \\ \text{often.} \end{array} \right\}$

cf. $\gamma_{\text{GE}} = 1$

Improve (i.e., decrease growth factor) by
doing COMPLETE PIVOTING

LU WITH COMPLETE PIVOTING: GECP

$A \rightarrow \dots \rightarrow$



Choose pivot from all of $A^{(k)}$
(instead of just the 1st column)

$$\Pi A \hat{\Pi} = LU$$

$\hat{\Pi}$ permutes the
column to do the full
pivoting

$$\gamma_{\text{GECP}} = \max_{\substack{1 \leq i \leq n \\ 1 \leq j \leq n \\ 1 \leq k \leq n}} \frac{|a_{ij}^{(k)}|}{|a_{ij}|} \leq \underbrace{C n^{\frac{1}{2}} n^{\frac{1}{4} \log n}}_{\text{Much better}}$$

• NOTE: GECP $\rightarrow$ $\boxed{\Pi A \hat{\Pi} = LU}$ = $\begin{bmatrix} L_1 & 0 \\ L_2 & \mathbb{1} \end{bmatrix} \begin{bmatrix} U_1 & U_2 \\ 0 & 0 \end{bmatrix}$

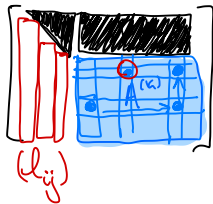
Full LU

(RANK-RETAINING FACTORIZATION) $\rightarrow$

$= \underbrace{\begin{bmatrix} L_1 \\ L_2 \end{bmatrix}}_{n \times r} \underbrace{\begin{bmatrix} U_1 & U_2 \end{bmatrix}}_{r \times n}$

Condensed LU

• REMARK: Look Pivoting (between GEPP and GECP)



"look" style search

PROBLEM FOR QUIZ 2: Find eigenvalues of finite difference matrix with bdy condition.

REMARK: Gauss - Jordan Elimination (UNSTABLE)

$A \rightarrow \dots \rightarrow U \rightarrow \dots \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 1 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 1 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$

Gives very large entries in L , which is bad.

(Reduced Row Echelon Form)

|||

(Hermite Form)

CHOLESKY DECOMPOSITION (POSITIVE DEFINITE MATRICES)

GOAL: Have $L = U^T$ in LU decomposition.

$$\boxed{A = R^T R} \Rightarrow x^T A x = x^T R^T R x = (R x)^T (R x) = \|R x\|^2 > 0$$

Unique if $r_{ii} > 0$
 $\forall i = 1, \dots, n$

not necessarily unique

FACT: $A = R^T R$ where $R \in GL(n)$ upper triangular
 $\Leftrightarrow A$ is symmetric positive definite

Pf: ($\Rightarrow$) Char, as above.

($\Leftarrow$) want to write $A = R^T R := F F^T$, $F := R^T$.

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{12} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} f_{11} & & & \\ f_{21} & f_{22} & & \\ \vdots & \vdots & \ddots & \\ f_{n1} & f_{n2} & \dots & f_{nn} \end{bmatrix} \begin{bmatrix} f_{11} & f_{21} & \dots & f_{n1} \\ & f_{22} & \dots & f_{n2} \\ & & \ddots & \vdots \\ & & & f_{nn} \end{bmatrix}$$

$$a_{11} = f_{11}^2 \rightsquigarrow f_{11} = \overset{\text{pick this}}{\oplus} \sqrt{a_{11}} = \sqrt{a_{11}}$$

$$a_{12} = f_{11} f_{21} = \sqrt{a_{11}} f_{21} \rightsquigarrow f_{21} = \frac{a_{12}}{\sqrt{a_{11}}}$$

$$a_{1i} = f_{11} f_{i1} \rightsquigarrow f_{i1} = \frac{a_{1i}}{\sqrt{a_{11}}}, i = 2, \dots, n$$

$$a_{22} = f_{21}^2 + f_{22}^2 \longrightarrow f_{22} = \sqrt{a_{22} - f_{21}^2} \in \mathbb{R} \quad \leftarrow \text{Sylvester's Thm}$$

⋮

keep comparing entries

⋮

FACT: Every diagonal entry of an A.S.O. is positive because

$$a_{kk} = e_k^T A e_k > 0$$

$$a_{kk} = f_{k1}^2 + f_{k2}^2 + \dots + f_{kn}^2 \Rightarrow f_{kj}^2 \leq a_{kk}$$

CHOLESKY IS VERY STABLE $\Leftarrow$ NO GROWTH FACTOR IN CHOLESKY

$$|f_{kj}| \leq \sqrt{a_{kk}}$$

$$f_{kk} = \sqrt{a_{kk} - \sum_{j=1}^{k-1} f_{kj}^2}, \quad f_{ik} = \frac{a_{ik} - \sum_{j=1}^{k-1} f_{kj} f_{ij}}{f_{kk}}$$

REMARK: SQUARE-ROOT-FREE CHOLESKY

$$A = LDL^T = \begin{bmatrix} 1 & & 0 \\ (l_{ij}) & \ddots & \\ & & 1 \end{bmatrix} \begin{bmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{bmatrix} \begin{bmatrix} 1 & & (l_{ij}) \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

REMARK: Another way to get Cholesky

$$A = FF^T = [f_1, \dots, f_n] \begin{bmatrix} f_1^T \\ \vdots \\ f_n^T \end{bmatrix} = f_1 f_1^T + \dots + f_n f_n^T$$

$$f_k = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ * \\ \vdots \\ * \end{bmatrix} \rightsquigarrow f_1 = \frac{1}{\sqrt{a_{11}}} a_1$$

So, rewrite

$$\begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & \boxed{A_2} \\ \vdots & \\ 0 & \end{bmatrix} = A^{(2)} := A - f_1 f_1^T = f_2 f_2^T + \dots + f_n f_n^T$$

Now,

$$A = S \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & \boxed{A_2} \\ \vdots & \\ 0 & \end{bmatrix} S^T \quad S := [f_1, e_2, \dots, e_n] = \begin{bmatrix} f_{11} & & & 0 \\ f_{21} & 1 & & \\ \vdots & & \ddots & \\ f_{n1} & 0 & & 1 \end{bmatrix}$$

$$\Rightarrow x \neq 0, \quad \begin{bmatrix} 0 \\ x \end{bmatrix}^T \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & \boxed{A_2} \\ \vdots & \\ 0 & \end{bmatrix} \begin{bmatrix} 0 \\ x \end{bmatrix} = y^T S^{-1} A S^{-T} y$$

$$\stackrel{(\dagger)}{=} x^T A_2 x = (S^{-T} y)^T A (S^{-T} y) > 0$$

$$\Rightarrow A_2 \succ 0.$$

$$\text{Thus, } f_2 = \frac{1}{\sqrt{a_{22}^{(2)}}} \begin{bmatrix} 0 \\ a_2^{(2)} \end{bmatrix}$$

$$A^{(3)} := A^{(2)} - f_2 f_2^T = \begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \boxed{A_3^{(3)}} & \\ 0 & 0 & & \end{bmatrix}$$

LECTURE 13

Solve $Ax_1 = b_1, \dots, Ax_p = b_p \rightsquigarrow AX = \overset{(1)}{B}$, $X = [x_1, \dots, x_p]_{n \times p}$
 e.g. compute A^{-1} ($AX = I$) $\Leftrightarrow (Ax_j = e_j)$ $B = [b_1, \dots, b_p]_{n \times p}$

Find: $A^{-1}B$. Solve (1) and multiply it to B with only one QR or LU or Cholesky, etc.

$\mathbb{R}^{n \times n} \ni A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \leftarrow$ Can generalize the decompositions from before to block matrices

• Block Elimination $M_{\perp} := I - U_{\perp} V_{\perp}^T$, $U_{\perp} = \begin{bmatrix} 0 \\ A_{21} A_{11}^{-1} \end{bmatrix}$, $V_{\perp} = \begin{bmatrix} I \\ 0 \end{bmatrix}$
 $\rightsquigarrow M_{\perp} A = \begin{bmatrix} I & 0 \\ -A_{21} A_{11}^{-1} & I \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ 0 & \underbrace{A_{22} - A_{21} A_{11}^{-1} A_{12}}_{\text{Schur Complement of } A_{11} \text{ in } A} \end{bmatrix}$
 $\Rightarrow L_{\perp} := M_{\perp}^{-1} = \begin{bmatrix} I & 0 \\ A_{21} A_{11}^{-1} & I \end{bmatrix} \rightsquigarrow \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} \\ 0 & U_{22} \end{bmatrix}$

Block Cholesky when $A \succ 0$, $A^T = A$

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} R_{11}^T & 0 \\ R_{12}^T & R_{22}^T \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{bmatrix}$$

$\leftarrow R_{11}, \dots, R_{22}$ don't need to be upper Δ !

Block LU $\left\{ \begin{array}{l} L_{11}, \dots, L_{22} \\ U_{11}, \dots, U_{22} \end{array} \right.$
 can be any matrices (not necessarily upper/lower Δ)

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \xrightarrow{LU} \boxed{\det A = (\det A_{11}) (\det A_{22} - A_{21} A_{11}^{-1} A_{12})}$$

SHERMAN - WOODBURY - MORRISON FORMULA: Consequence of LDU

$$(A + BC)^{-1} = A^{-1} - A^{-1}B(I_d + CA^{-1}B)^{-1}CA^{-1}.$$

MEMORIZE ALL

1) Block LU: A_{11} invertible,

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} I_d & 0 \\ A_{21}A_{11}^{-1} & I_d \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ 0 & S \end{bmatrix}, \quad S := A_{22} - A_{21}A_{11}^{-1}A_{12}$$

(Schur Complement
of A_{11} in A)

2) Block LDU: A_{11} invertible

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} I_d & 0 \\ A_{21}A_{11}^{-1} & I_d \end{bmatrix} \begin{bmatrix} A_{11} & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} I_d & A_{11}^{-1}A_{12} \\ 0 & I_d \end{bmatrix}$$

3) Block Cholesky: $A = A^T > 0$,

$$0 < \begin{bmatrix} A_{11} & A_{21}^T \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} R_{11}^T & 0 \\ R_{12}^T & R_{22}^T \end{bmatrix} \begin{bmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{bmatrix}$$

$$= \begin{bmatrix} R_{11}^T R_{11} & R_{11}^T R_{12} \\ R_{12}^T R_{11} & R_{12}^T R_{12} + R_{22}^T R_{22} \end{bmatrix}.$$

4) Block Inverse: $A, S := D - CA^{-1}B$ invertible,

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} + A^{-1}S^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{bmatrix}$$

LECTURE 14

ITERATIVE METHODS

Nov 19th, 2025

GOAL: Find x s.t. $Ax = b \leadsto \lim_{k \rightarrow \infty} x^{(k)} = x_* = A^{-1}b$.

1) SPLITTING METHODS: $A = M - N$

$$(M - N)x = b \leadsto Mx = Nx + b$$

$$x^{(0)}, \text{ for } k=1, 2, 3, \dots, \quad Mx^{(k+1)} = Nx^{(k)} + b$$

$\uparrow$ easy to invert

2) Semi-iterative methods (next time)

3) Krylov Subspace Methods

REMARK: Cayley-Hamilton $\leadsto A \in GL(n, \mathbb{R})$
 $p(\lambda) \in \mathbb{R}[\lambda]$

$$p(A) = a_0 \mathbb{I} + a_1 A + a_2 A^2 + \dots + a_d A^d = 0$$

$$\Rightarrow A^{-1} = -\frac{a_1}{a_0} \mathbb{I} - \frac{a_2}{a_0} A - \dots - \frac{a_d}{a_0} A^{d-1}$$

$$x_* := A^{-1}b = -\frac{a_1}{a_0} b - \frac{a_2}{a_0} Ab - \dots - \frac{a_d}{a_0} A^{d-1}b$$

$$\Rightarrow x_* = A^{-1}b \in \text{span} \left\{ b, Ab, \dots, A^{d-1}b \right\}$$

KRYLOV
SUBSPACE $\equiv K_d(A, b)$

$$k = \# \{ \text{distinct ew's of } A \} \leadsto x^{(k)} = x_* = A^{-1}b$$

1) SPLITTING METHOD: $A := M - N$, $A \in GL(n, \mathbb{R})$

$$\Rightarrow Mx^{(k+1)} = Nx^{(k)} + b$$

Error: $e^{(k)} = x^{(k)} - \underbrace{x_*}_{\leftarrow x_* = A^{-1}b \Rightarrow Mx_* = Nx_* + b}$

Thus: $M(x^{(k+1)} - x_*) = N(x^{(k)} - x_*)$

$$\rightsquigarrow e^{(k+1)} = \underbrace{M^{-1}N}_{\substack{=: \\ B}} e^{(k)} \rightsquigarrow \boxed{e^{(k+1)} = B e^{(k)}}$$

Take: $\|\cdot\|$ consistent (e.g., ∞) $\Rightarrow \|e^{(k+1)}\| \leq \|B\| \|e^{(k)}\|$

By induction: $e^{(k+1)} = B e^{(k)} = B^2 e^{(k-1)} = \dots = B^{k+1} e^{(0)}$

$$\Rightarrow \|e^{(k+1)}\| \leq \|B^{k+1}\| \|e^{(0)}\| \leq \|B\|^{k+1} \|e^{(0)}\|.$$

THM: $\lim_{k \rightarrow \infty} x^{(k)} \stackrel{\text{def}}{=} \lim_{k \rightarrow \infty} B^k e^{(0)} = 0 \quad \forall e^{(0)} \in \mathbb{R}^n$

$$\lim_{k \rightarrow \infty} B^k = 0$$

$$\begin{array}{c} \Updownarrow \\ \text{Jordan} \\ \text{Canonical} \\ \text{Form} \end{array}$$

$$\rho(B) < 1$$

CONVERGENCE RATES

sublinear: $\frac{1}{n} \rightarrow 0$, linear: $\frac{1}{2^n} \rightarrow 0$, quadratic: $\frac{1}{2^{2^n}} \rightarrow 0$

linear = "exponential"

DEF:

RATE OF CONVERGENCE $\equiv \limsup_{k \rightarrow \infty} \frac{\|e^{(k+1)}\|}{\|e^{(k)}\|^p}$

$\left\{ \begin{array}{l} p=1 \text{ (linear)} \\ p=2 \text{ (quadratic)} \end{array} \right.$

PSEUDO-RATE OF CONVERGENCE $\equiv \|B\| \leftarrow \text{b/c } \|e^{(k+1)}\| \leq \|B\| \|e^{(k)}\|$

PSEUDO-PSEUDO RATE OF CONVERGENCE $\equiv \rho(B)$

AVERAGE RATE OF CONVERGENCE OVER k -STEPS $\equiv \|B^k\|^{1/k}$

$\lim_{k \rightarrow \infty} \|B^k\|^{1/k} = \rho(B)$

* JACOBI METHOD:

$A = \begin{bmatrix} \text{lower triangular} \\ L \end{bmatrix} + \begin{bmatrix} \text{diagonal} \\ D \end{bmatrix} + \begin{bmatrix} \text{upper triangular} \\ U \end{bmatrix}$

$-N$

$M = D, N = -(L+U) \Rightarrow Mx^{(k)} = (L+U)x^{(k)} + b$

Jacobi doesn't work if $D \equiv 0 \dots$ so it's only good strictly diag. dom. matrices (e.g. 7.0) $\rightarrow B_{\text{Jacobi}} = D^{-1}(L+U)$

THM: If A is strictly diagonally dominant, then $\|B_J\| < 1$

ALGORITHM: (JACOBI METHOD) For $j=1, \dots, n$, ← easily parallelizable

$$x_j^{(k+1)} := \frac{1}{a_{jj}} \left[b_j - \sum_{l=1}^{j-1} a_{jl} x_l^{(k)} - \sum_{l=j+1}^n a_{jl} x_l^{(k)} \right], \quad k=1, 2, 3, \dots$$

* GAUSS - SEIDEL

$$A = \underbrace{\begin{bmatrix} \text{lower triangle} \end{bmatrix}}_{L} + \underbrace{\begin{bmatrix} * & & \\ & * & \\ & & \ddots \\ & & & * \end{bmatrix}}_{D} + \underbrace{\begin{bmatrix} \text{upper triangle} \end{bmatrix}}_{U}$$

$$\Rightarrow (L+D)x^{(k+1)} = -Ux^{(k)} + b$$

$$B_{GS} = -(L+D)^{-1}U$$

THM: If A is strictly diagonally dominant, then $\|B_{GS}\| < 1$

ALGORITHM: (GAUSS - SEIDEL) For $j=1, \dots, n$,

$$x_j^{(k+1)} = \frac{1}{a_{jj}} \left[b_j - \sum_{l=1}^{j-1} a_{jl} \underbrace{x_l^{(k+1)}}_{\text{of. Jacobi}} - \sum_{l=j+1}^n a_{jl} x_l^{(k)} \right]$$

← not parallelizable

← Twice as fast than Jacobi

THM: Under mild-conditions $\rho(B_{GS}) = \rho(B_J)^2$

—————→—————

* SOR

ALGORITHM (SOR): For $j=1, 2, \dots, n$,

$$x_j^{(k+1)} = \frac{\omega}{a_{jj}} \left[b_j - \sum_{l=1}^{j-1} a_{lj} \overset{gs}{x_j^{(k+1)}} - \sum_{l=j+1}^n a_{lj} x_j^{(k)} \right] + \underbrace{(1-\omega) x_j^{(k)}}_{\text{previous vec. contains useful info.}}$$

$0 < \omega < 2$ (EXTRAPOLATION)

$$\omega = \frac{2}{1 + \sqrt{1 + \rho(B_J)^2}} \rightsquigarrow \boxed{B_\omega = \left(\frac{1}{\omega} D + L \right)^{-1} \left[\left(\frac{1}{\omega} - 1 \right) D - U \right]}$$